

Executive Summary: Revenue Control Guide for Vital Records Data

This guide outlines a step-by-step strategy to control and maximize revenue generated from vital records (birth and de

1. Map Existing Revenue Streams:

- Direct sales (certificates)
- Verification services (EVVE, EVVE FOD)
- Federal contracts (SSA, CMS, VA)
- Research access (NDI, custom data sets)
- Commercial use (banks, insurers, genealogy, title companies)

2. Establish Cost Accounting Framework:

- Track transaction costs (certificate issuance, verification)
- Categorize into administrative, technology, and compliance

3. Implement Revenue Controls:

- Standardize fee schedules across states
- Tiered pricing (citizen, institutional, commercial)

4. Optimize Federal and State Partnerships:

- Negotiate multi-year contracts with SSA/CMS
- Leverage federal cost-sharing for modernization

5. Expand Data-as-a-Service (DaaS):

- Real-time API verification subscriptions
- Tiered researcher access (public-use vs restricted)

6. Strengthen Fraud Prevention:

- Certified access through LADMF
- Monitor fraudulent requests to reduce leakage

7. Reinvest Revenue into Modernization:

- Upgrade EBRs/EDRS systems
- Enhance cybersecurity & staff training

8. Monitoring & Evaluation:

- KPIs: cost recovery, revenue per record, income breakdown
- Quarterly audits & annual reviews

Result: A financially sustainable system that maximizes revenue, reduces fraud, and supports modernization of the na

Department of Health and Human Services

**OFFICE OF
INSPECTOR GENERAL**

**BIRTH CERTIFICATE
FRAUD**



**JUNE GIBBS BROWN
Inspector General**

**SEPTEMBER 2000
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OFFICE OF INSPECTOR GENERAL

The mission of the Office of Inspector General (OIG), mandated by Public Law 95-452, as amended by Public Law 100-504, is to protect the integrity of the Department of Health and Human Services programs as well as the health and welfare of beneficiaries served by them. This statutory mission is carried out through a nationwide program of audits, investigations, inspections, sanctions, and fraud alerts. The Inspector General informs the Secretary of program and management problems and recommends legislative, regulatory, and operational approaches to correct them.

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EXECUTIVE SUMMARY

PURPOSE

To provide an update on the nature and extent of birth certificate fraud.

BACKGROUND

Legitimate birth certificates provide vital information about the person whose name appears on the certificate. Issuing birth certificates is the responsibility of State vital records registrars and numerous local issuing entities. While originally intended for the sole purpose of birth registration, birth certificates are now used extensively for employment purposes and to obtain benefits or other documents used for identification.

Over the last 25 years, a number of studies have addressed problems related to false identification and the misuse of birth certificates. These studies conclude that false identification is a major factor in crime, and that most, if not all, Federal fugitives and drug trafficking crimes are associated with false identification. They also conclude that stolen, counterfeit, and altered birth certificates are often used as “breeder documents” that allow the holder to obtain documents needed to create new identities. The Office of Inspector General has conducted three inspections focused specifically on birth certificate fraud that identified a number of vulnerabilities in birth certificate processes. Because so many Federal and State agencies rely on birth certificates to assist them in determining eligibility for services and benefits, it is important that these agencies have current information on the nature and extent of birth certificate fraud to assist them in the proper assignment and protection of benefits.

This inspection was initiated at the request of the Department of Health and Human Services to provide information which the Department could use in responding to responsibilities placed on it as a result of the Illegal Immigration Reform and Immigrant Responsibility Act of 1996.

We used two mechanisms to gather information. We gathered general information regarding vital records policies and procedures from 53 primary vital records registrars using mail surveys. We also collected more in-depth information through personal interviews with State registrars, fraud investigators, and local and State vital records, Passport Services, Immigration and Naturalization Services, Social Security Administration, and public assistance staff in five States and New York City. We also met with the Center for Disease Control and Prevention (CDC), National Center for Health Statistics, staff who provided us with comments on the draft of this report.

FINDINGS

Fundamental, Irreconcilable Conflicts Surround Birth Certificate Purposes and Uses

A certified copy of a birth certificate is proof only that a birth occurred and was recorded. For that purpose, it may be desirable that the public be allowed easy access to them. However, the agencies and organizations that use birth certificates as proof of

identification for employment purposes, to obtain benefits or other documents (e.g., driver's licenses, Social Security cards, and passports), and to assist them in determining eligibility for public assistance and other benefits, may have concerns with how easily certified copies of birth certificates can be obtained. These conflicting perspectives are at the very heart of the birth certificate controversy.

Birth Certificates Continue to be Used as “Breeder Documents” and are Easy to Obtain

Virtually all Federal and State agencies agree that fraudulent birth certificates are used as “breeder documents” to obtain the genuine documents needed to create new identities, and that fraudulent birth certificates are easy to obtain. Factors which contribute to their use as “breeder documents” include the following:

- currently, *6,422 different entities issue birth certificates*. This large number of State, county, city, township, and other entities that issue birth certificates increases opportunities for fraud, theft, bribery, and other methods of illegally obtaining birth certificates;
- *thirteen States allow “open” access* to birth records, which allows virtually anyone to purchase copies of any birth certificates on file; and
- *birth certificates can be purchased without identification* from some vital records offices and issuing entities.

Birth Certificate Fraud is Hard to Detect

Many altered or counterfeit birth certificates and genuine birth certificates held by imposters may go undetected. The reasons why these fraudulent birth certificates are hard to detect include the following:

- *over 14,000 different versions of birth certificates are in circulation;*
- *nearly 4 million United States births were registered in 1999;*
- *security features* contained in the paper used to issue birth certificates, as well as formats and signatures, vary among State vital records offices and the many local entities issuing them;
- *technological advances* in the Internet, scanners, color printers, and copiers make it easier to obtain genuine birth certificates and create counterfeit ones;
- between 85 and 90 percent of the birth certificate fraud encountered by the Immigration and Naturalization Services and Passport Services staff is the result of *genuine birth certificates held by imposters* -- the most difficult fraud to detect; and
- Federal and State agency staff report receiving only *limited training* focused on the detection of fraudulent birth certificates.

State Practices Create Opportunities for Fraud

It was the consensus of those we interviewed that a number of State practices create opportunities for fraud. Those practices include the following:

- *delayed, amended, and midwife birth registrations* that are based on affidavits of personal knowledge, include no documentary evidence, and are not often marked or overlaid accordingly;
- *delays in matching death and birth records* can make the identities of many deceased persons easy to assume between the time the person dies and the time the death and birth records are matched;
- *questionable physical security* situations that create opportunities for fraud; and
- *limited oversight* of local issuing entities by State vital records offices.

Birth Certificate Fraud is Seldom Prosecuted

Virtually all of the Federal and State agency staff we talked with indicate birth certificate fraud is seldom prosecuted unless it can be linked to large dollar losses or other punishable crimes. Most staff also indicate that many prosecutors are reluctant, or refuse to take birth certificate fraud cases in which the only charge is attempting to obtain another individual's birth certificate, or counterfeiting or altering a birth certificate. At the same time, misconceptions exist surrounding the security and integrity of birth certificates.

CONCLUSIONS

Birth Certificates Alone do not Provide Conclusive or Reliable Proof of Identity

Many agencies and organizations request that individuals provide their birth certificates to receive a benefit or service, or to support the issuance of other documents often used for identity purposes (e.g., driver's license). However, agencies who rely on birth certificates as a means of establishing identity must understand the limitations of accepting a birth certificate as proof of age, citizenship, or identity. For example, genuine documents obtained with counterfeit birth certificates can be used to obtain genuine birth certificates. Thus, it is inherently illogical to require someone to prove their identity using potentially fraudulent identity documents spawned by false birth certificates in order to obtain a birth certificate.

Further, it would be Impractical to Redesign Birth Certificates to Make them Reliable Identification Documents in and of Themselves

Efforts to make the birth certificate into a reliable identity document are complicated by the more than 14,000 different legitimate versions in existence, and the more than 6,000 entities which issue them and the processes they use to do so. Efforts are also complicated by the ease with which birth certificates can legitimately be obtained and counterfeited, and the fact that the majority of fraud is now being committed by imposters

using genuine birth certificates. Also, any changes to the birth certificate itself will take essentially a lifetime to become effective.

Some Efforts to Redesign Birth Certificates Might Even be Undesirable

The primary purpose for which birth certificates were created -- to document and record births -- is served well by the large number of entities that issue them and the technology which makes them readily and quickly available. Because redesigning birth certificates could jeopardize their availability, to do so might be undesirable. Unfortunately, that availability contributes to fraud and the unreliability of birth certificates as identification documents.

Nevertheless, Since Birth Certificates can Play an Important Role in Establishing Identity, Their Integrity Should be Improved

When used in combination with other documents, birth certificates can add to the level of proof in establishing eligibility and identity. As noted previously, many agencies use them in this way. Therefore, it is important that the processes used to issue birth certificates be standardized and recent advances in technology utilized to ensure birth certificate integrity parallels that of other identification documents. It is also important that user agencies be vigilant in their detection of fraudulent documents and documents held by imposters. We have included consensus suggestions for improving the birth certificate process and the detection of fraud in the body of the report.

In Addition, Federal and State Program Administrators Should Assess the Proofs of Identity They Will Accept

Even if their security is improved, birth certificates may still not be the best proof of identity. For this reason, program administrators may not want to use birth certificates at all, or use them only with other documents, as noted above. Agencies need to specify documents and methods of proving identity (e.g., fingerprints, testimony of relatives) they will accept in determining eligibility for services. Given what we have learned, if program administrators continue to include birth certificates in the proofs of identity they will accept, they should also reconsider what steps they will take to detect fraudulent certificates and to secure valid ones.

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INTRODUCTION

PURPOSE

To provide an update on the nature and extent of birth certificate fraud.

BACKGROUND

The Birth Certificate Process

Legitimate birth certificates provide vital information about the person whose name appears on the certificate (i.e., legal proof of parentage, citizenship, date, place, and time of birth). While originally intended for those purposes, their role has evolved to the point where they are now used extensively for employment purposes and to obtain benefits or other documents (e.g., driver's licenses, Social Security cards, passports, State identification documents). The issuance of birth certificates in the United States is the responsibility of 57 State vital records registrars and numerous other entities assigned vital records responsibility.

No Federal requirements exist regarding the reporting and collection of birth certificate information. However, the 1992 *Model State Vital Statistics Act and Regulations* published by the National Center for Health Statistics within the Centers for Disease Control and Prevention (CDC) state that "Even though the legal responsibility for the registration of vital records rests with the individual States, the States and the National Center for Health Statistics (the Federal partner) work together to build a uniform system that produces records to satisfy the legal requirements of individuals and their families and also to meet statistical and research needs at the local, State, and national levels. The cooperation includes the development and promotion of standard certificates and reporting forms, training and quality control programs, and model legislation." The information contained on birth certificates is the basis for the vital record information reported to the National Center for Health Statistics by State vital records registrars. States report vital records information without names to the National Center for Health Statistics for research and statistical purposes.

Previous Reports on Birth Certificate Fraud

Federal Advisory Committee on False Identification - Problems related to false identification and misuse of birth certificates have been addressed in a number of studies conducted over the last 25 years. In 1974, the Federal Advisory Committee on False Identification (FACFI), commissioned by the Attorney General of the United States, recognized the criminal use of false identification documents. In its 1976 report entitled

The Criminal Use of False Identification, the committee concluded that 100 percent of all Federal fugitives and 80 percent of all drug trafficking are associated with false identification. They also reported that false identification is a major factor in crime, including illegal immigration and flight from justice, and that falsified or stolen vital statistics (i.e., birth certificates) are used as “breeder documents.” (“Breeder documents” refer to documents that allow the holder to obtain other documents -- passports, driver’s licenses, etc. -- and benefits, such as resident status, Social Security benefits, loans, and other government aid, including Temporary Assistance to Needy Families, Food Stamps, and Medicaid). The Federal Advisory Committee also issued a supplemental report entitled *A Plan for Reducing the Abuse of Birth Certification*. In 1984, the Laws at Work Task Force, co-chaired by top HHS officials, issued a report entitled *A Report of the Task Force on Criminal Implications of False Identification*. These two reports on false identification have findings similar to the 1976 report.

National Association for Vital Records - In 1995, the National Association for Vital Records and Health Statistics (now called the National Association for Public Health Statistics and Information Systems), issued a staff report identifying many of the same problems associated with birth certificate fraud. They recommended uniformity and standardization of the forms and paper used to issue certified copies of birth certificates and further suggested that more resources be directed toward matching death and birth information.

Office of Inspector General - The Office of Inspector General (OIG) conducted two inspections focused specifically on birth certificate fraud (*Birth Certificate Fraud*, OAI-86-02-00001, March 1988, and *Birth Certificate Fraud Update - A Management Advisory Report*, OEI-02-91-01530, December 1991). The OIG also issued a third report (*Citizenship and Alien Verification - Information*, September 1996) in which birth certificates were discussed. The first inspection identified vulnerabilities in the birth certificate process, the many forms of birth certificates in existence, and issuance procedures. The report recommended both interstate and intrastate standardization of birth certificates, minimum security standards, and improvements in matching death and birth records, and further recommended that States work cooperatively with the Social Security Administration (SSA) to establish procedures to issue Social Security numbers to infants at birth. The second inspection was conducted in response to a request from the Commissioner of the Social Security Administration and summarized SSA’s efforts to control birth certificate fraud, which included verification of United States Virgin Island birth certificates and assignment of Social Security numbers at birth. In this study, we found the nature and extent of birth certificate fraud relatively unchanged since 1988 and that major weaknesses continued to hamper the reliability of birth certificates as evidence of eligibility for program services and benefits, even though some incremental improvements had been made. We reported the same problems and weaknesses in our 1996 report.

This Inquiry

This inspection was initiated at the request of the Department of Health and Human Services in the fall of 1999. The Department requested that the OIG provide information that the Department could use in responding to responsibilities placed on it as a result of the Illegal Immigration Reform and Immigrant Responsibility Act of 1996. More specifically, the Act requires the Secretary to “submit a report to Congress on ways to reduce the fraudulent obtaining and use of birth certificates.” This inspection focuses on the processes of issuing and fraud associated with certified copies of birth certificates. The OIG was asked to provide this information because of our previous work in the area of birth certificate fraud at a time when the SSA was located within the Department, and because public assistance programs currently funded by the Department still rely on birth certificates as one proof of identification in determining eligibility. Further, because so many State and Federal agencies rely on birth certificates in determining eligibility for services and benefits, we feel it is important that they have current information regarding the status of birth certificate fraud.

METHODOLOGY

We used two mechanisms to conduct this inspection. We first solicited information from the 57 primary vital records offices (50 States, Guam, Puerto Rico, United States Virgin Islands, American Samoa, Commonwealth of the Northern Mariana Islands, New York City, and Washington, DC) using mail surveys. This enabled us to quickly gather general information regarding vital records policies and procedures; security measures in place to protect against birth certificate fraud; coordination with other agencies; and enforcement penalties for committing identity fraud. Of the 57 surveys mailed, 53 were completed and returned. We received completed surveys from vital records registrars in each of the 50 States, and New York City, Washington, DC, and Puerto Rico. We did not receive completed surveys from American Samoa, Commonwealth of the Northern Mariana Islands, Guam, or the United States Virgin Islands.

We selected six locations -- five States (California, Florida, Massachusetts, New Mexico, and Texas) and New York City -- for additional data collection. Of these States, 4 were among the 11 States included in the 1988 and 1991 OIG inspections, as was New York City. The original 11 sites were selected because they represent states with large cities, cities that serve as major ports, or cities along the United States-Mexico border where birth certificate fraud was more concentrated. Because we believe this premise remains valid, we included some of these same States in this study. We included New Mexico based on information obtained from State registrars, the Immigration and Naturalization Service, and the National Center for Health Statistics during preinspection. The locations selected represent a mix of States with “open” and “restricted” (limited) public access to records, and include various geographic regions.

In the six locations, we collected more in-depth information about the coordination between agencies, matching death registration and birth certificates, security measures, enforcement, and the effects of new technology on birth certificate fraud. This information was obtained through personal interviews with State registrars, fraud investigators, local and State vital records agency staff, and local and State Temporary Assistance for Needy Families agency staff. We also collected information from the Departments of Motor Vehicles and Food Stamps staffs who require birth certificates as proof of eligibility for program services or benefits, and officials from SSA, Immigration and Naturalization Service, and State Department's Passport Services. We also met with the Center for Disease Control and Prevention, National Center for Health Statistics staff who provided us with comments on the draft of this report

We conducted this inspection in accordance with the ***Quality Standards for Inspections*** issued by the President's Council on Integrity and Efficiency.

FINDINGS

In conducting this inspection, we heard many stories of people whose lives have been adversely affected by birth certificate fraud -- people who were personally victimized by stolen identities and suffered financial ruin. Others stories were more far-reaching, describing the criminal use of fraudulent birth certificates to evade law enforcement for crimes already committed, commit bank or credit card fraud, or obtain services and benefits for which individuals were not entitled. We also heard stories about fraudulent birth certificates used by terrorists and drug traffickers to acquire passports for international travel, and by illegal aliens attempting to avoid detection and deportation. The common threads running through each of these stories are 1) fraudulent birth certificates were involved, 2) most false identity crimes go undetected, 3) once detected, few false identity crimes are successfully prosecuted, and ultimately, 4) each one of us is affected by birth certificate fraud every day.

Recent Congressional and media attention on the subjects of identity theft and the sale of false identification documents via the Internet, and the statements of State and Federal staff with whom we spoke, reflect concerns regarding the fraudulent use of birth certificates. Virtually everyone with whom we spoke indicated that they believe birth certificate fraud is increasing, and 34 State registrars responded in the survey that birth certificate fraud has increased in the last 10 years.

To this end, the primary focus of this report and the majority of our findings address the status of birth certificate fraud. However, one of our most striking findings is related to the disconnect between the intended purposes of birth certificates and the purposes for which they are used.

Fundamental, Irreconcilable Conflicts Surround Birth Certificate Purposes and Uses

A Birth Certificate is Proof Only that a Birth Occurred and was Recorded. It is important to recall the intended purpose of a birth certificate -- to certify that a birth was documented and recorded. For that purpose, it is not unreasonable that State laws allow public access to these records. Efforts to restrict access to birth certificates would diminish their value for the purpose they were originally designed to serve.

However, Birth Certificates are Used as Proof of Age, Citizenship, and Identity. Birth certificates are widely recognized as proof of age, place of birth, and identity. They are also used extensively for employment purposes, to obtain benefits or other documents (e.g., driver's licenses, Social Security cards, and passports), to assist in determining eligibility for public assistance and other benefits, to enroll children in school, and as proof of age eligibility for sports and other age restricted activities. However, because

they were never designed to provide sole proof of identity, and because a birth certificate cannot be positively linked with an individual, their use for that purpose is questionable.

Concerns about Birth Certificates are Largely the Result of Conflicting Perspectives. The key to understanding the conflicts between the goals of State and local entities who register births and issue birth certificates and agencies and organizations that rely on birth certificates as proof of identification is understanding and interpreting individual perspectives. For example, those seeking access to public records may look favorably on vital records offices' efforts to make it easier to request and obtain birth certificates, while those using birth certificates for identity purposes may have concerns with how easily certified copies of birth certificates can be obtained.

These varying perspectives are at the very heart of the birth certificate controversy and should be kept in mind as you read the following findings related to birth certificate fraud.

Birth Certificates Continue to be Used as “Breeder Documents” and Are Easy to Obtain

As we previously reported in 1988, 1991, and 1996, birth certificates continue to be used as “breeder documents” from which other supporting documents can be secured to alter identities and fraudulently obtain services and benefits. Virtually all Federal and State agencies agree that fraudulent birth certificates are used to obtain genuine documents, and in concert with other fraudulent documents, to create new identities. The perpetrator usually begins with a purchased, stolen, counterfeit, or altered birth certificate. The birth certificate is then used as the basic evidence of age, citizenship, and identity to seek and obtain other documents and/or benefits.

Birth Certificate → Social Security Card → Driver's License
= Services/Benefits/Credit/Crime

The following two instances demonstrate how birth certificates have provided the basis for improper program payments.

A joint investigation conducted by the Assistant U.S. Attorney for the Southern District of Texas in cooperation with the Immigration and Naturalization Services, Social Security Administration, public assistance, Food Stamps, U.S. Postal Service, and Secret Service identified 100 cases in which fraudulent birth certificates had been used to obtain public assistance, Food Stamps, and Medicaid benefits totaling \$514,741.¹

¹ Source: Texas Department of Human Services, Office of Inspector General

In another case, fraudulent birth certificates found at the scene of an automobile accident were linked to \$1 million in public assistance and Food Stamps fraud. Some of the birth certificates used to obtain these services were genuine documents that had been sold by the persons whose names appeared on them. Others were counterfeit documents printed on genuine birth certificate banknote paper used by one State's vital records office, which investigators suspect were acquired as "part of an inside job." Staff interviewed in a public assistance agency referred to the children created to obtain these services as "paper babies" because they exist only on paper, for the sole purpose of fraudulently obtaining benefits.²

The first step in understanding why birth certificates are easy to obtain, thus making them useful as "breeder documents," is to understand how they are issued. It is also important to know that the issuance of birth certificates varies from State to State and can vary within States. Factors which contribute to the continued use of birth certificates as "breeder documents," and that remain relatively unchanged since our earlier reports are as follows.

Currently, 6,422 Entities Issue Birth Certificates. The majority of States use a decentralized approach for issuing birth certificates, relying on local offices over which they have limited control. In 39 States and Puerto Rico, birth certificates are issued by a number of local vital records registrars and other officials (e.g., county clerks, justices of the peace, mayors) in county, city, and township offices in addition to the State vital records office. Only 11 States, Washington, DC, and New York City use a centralized operation for issuing birth certificates.

The large number of locations issuing birth certificates potentially increases the opportunities for fraud, theft, bribery, or other methods to illegally obtain birth certificates. In total, the registrars responding to our survey report 6,417 different locations where birth certificates can be issued. The National Center for Health Statistics indicates an additional five entities issue birth certificates within the jurisdictions of the four primary vital registrars that did not respond to our survey. A chart outlining the reported number of locations where birth certificates are issued in each State, New York City, Washington, DC, and Puerto Rico is located in Appendix A.

Most State and county issuing offices are located within the auspices of their respective health departments. The sale of birth certificates serves as source of revenue for States and the only source of revenue for some State health departments, thus creating no disincentive for issuing birth certificates. The sale of birth certificates can serve also as a major source of revenue for many local offices. For these departments and offices, restrictions on the issuance of birth certificates could adversely affect revenues.

² Source: Kansas Department of Health and Environment, Center for Health and Environmental Statistics, Office of Vital Statistics

Some States Allow “Open” Access to Birth Records. In 36 States, New York City, Washington, DC, and Puerto Rico, access to birth records is considered “restricted.” In these locations only the person listed on the birth certificate and others designated for access by State or local governments (e.g., parent, child, sibling, grandparent, or others who demonstrate a direct and tangible interest) can obtain certified copies of a birth certificate. However, in 14 States, public access to birth records is “open” at the State or local level, and virtually anyone can review birth records or purchase a copy of any birth certificate from issuing entities as long as they know the name and birth date of the person listed on the birth certificate. There are 2,375 such “open” access locations in these States, which account for 37 percent of the 6,422 issuing offices nationwide.

- Ten of the 14 States allow “open” access at both the State and local levels and treat requests for birth certificates the same in all locations.
- Two States allow “open” access at the State level, but allows only “restricted” access in all local offices within the State.
- Two States allow only “restricted” public access to birth records at the State level, but “open” at the local level. They allow birth registration indexes and/or actual birth certificates to be openly viewed in local offices and thus provide enough information to request copies of certificates from the State registrar’s office.

We have identified “open” access at both the State and local level, and the number of issuing offices in each of those States, in the following chart.

Open Access to Vital Records

<u>State</u>	<u>Issuing Entities</u>		<u>Total Number of Issuing Offices</u>
	<u>State Level</u>	<u>Local Level</u>	
California	✓	✓	63
Iowa		✓	101
Kentucky	✓		1
Maine	✓	✓	498
Massachusetts	✓	✓	352
Minnesota	✓	✓	98
New Jersey	✓	✓	566
North Carolina		✓	102
Ohio	✓	✓	144
South Dakota	✓	✓	65
Tennessee	✓		28
Vermont	✓	✓	248
Washington	✓	✓	34
Wisconsin	✓	✓	75
			<u>2,375</u>

Copies of birth certificates can be obtained from primary vital records offices and local entities by using a variety of methods. With the exception of California, all of the responding primary vital records registrars indicate birth certificates can be obtained from their offices on a “walk-in” basis. All 53 State vital records offices and local offices in 34 States also issue birth certificates in response to requests they receive in the mail. Thirty-eight primary offices and an unknown number of local offices in seven States also allow credit card purchases of birth certificates based on telephone requests. Birth certificates can also be purchased through the Internet in 29 States, and from an unknown number of local offices.

Proof of Identification Is Not Always Required to Obtain Copies of Birth

Certificates. Proof of identification is required for walk-in requests in only 30 primary vital records offices and local offices in 19 States. In addition, only 11 States require persons to provide proof of identification when requesting copies of birth certificates by mail. Local offices in fewer States (7) require proof of identification for mail requests. Even fewer primary and local offices request proof of identification for Internet and facsimile requests as indicated in the following chart.

Number of Vital Records Offices Accepting Type of Request and Proof of Identification Required

<u>Type of Request</u>	<u>State Accept/Require ID</u>	<u>Local Accept/Require ID</u>
Walk-in	52/30	39/19
Phone/Credit Card	38/7	10/1
Mail/Check or Money Order	53/11	34/7
Mail/Credit Card	22/2	8/2
Internet/Credit Card	29/5	3/2
Facsimile/Credit Card	41/9	10/2

When asked what they consider acceptable forms of identification, the registrars interviewed said they would prefer a driver’s license or State issued identification card as proof of identity, but some indicated they would also accept identification documents with “any type of control number” (e.g., employment badges, library cards, bus passes) or utility bills as proof of identification. We also learned the laws in one “open access” State (California) prohibit issuing offices from asking for identification. As long as the person requesting the birth certificate can provide the required information (i.e., name, date of birth), he or she can obtain a certified copy of a birth certificate. In addition to the lack of identification documents, 43 States also have no limits on the number of copies of birth certificates that can be purchased, and 31 States have no restrictions on the age of persons to whom they will sell birth certificates. The lack of restrictions make it easier to perpetrate systematic fraud.

Fraudulent Birth Certificates are Hard to Detect

With the exception of detection by highly trained Immigration and Naturalization Services and Passport Services staff, information we obtained indicates many altered or counterfeit birth certificates and genuine birth certificates held by imposters may go undetected. Officials who use birth certificates to determine eligibility for services and benefits say they fail to detect many fraudulent documents, and the ones they do detect are “only the tip of the iceberg.” In attempting to identify fraudulent birth certificates, SSA, public assistance, Department of Motor Vehicles, and some Immigration and Naturalization Service staff indicate their efforts to detect fraudulent birth certificates focus only on obvious alterations, such as erasures, smudges, white-out, misspelled words, offset margins, poor seals, dates that do not match, or photocopies. However, information we obtained indicates birth certificates are vulnerable to fraud beyond the obvious in the following areas.

The Large Number of Legitimate Birth Certificates in Circulation Make Fraud Detection Difficult. Staff at the Immigration and Naturalization Service’s Forensics Document Laboratory³ estimate more than 14,000 different versions of legitimate birth certificates currently exist. The number of different versions is the result of the more than 6,000 entities issuing birth certificates using different formats, types of paper, and different signatures (e.g., State registrars, county registrars, or clerks, mayors, and justices of the peace). Once a birth certificate is issued, it never expires, meaning that security features added to new birth certificates will offer no protection against fraud in previously issued birth certificates. The National Center for Health Statistics reports that 3,957,829 United States births were registered in 1999.

In addition, responses to our survey revealed State vital records offices currently issue 113 different types of certified copies of birth records. This number does not account for the number of variations in local office issuance. Fifty-one of the 53 primary vital records offices issue certified photocopies of actual birth records, 37 issue certified copies of computerized abstracts of birth records, 17 issue wallet-sized birth certificates/cards, and 8 issue commemorative birth certificates, each with their own unique security features and signatures. In addition, survey respondents report that, in 20 States, local entities issue full photocopies of actual birth records, 16 States’ local entities issue certified copies from computerized abstracts, 17 States’ local entities issue wallet-sized certificates, and 4 States’ local entities can issue commemorative birth certificates.

Differences in Paper, Security Features, Formats, and Signatures Make Fraudulent Birth Certificates Hard to Detect. All State vital records offices issue birth certificates on security paper, but the security features vary from State-to-State. Some local offices

³ The Forensics Document Laboratory provides a wide variety of forensic document analysis and law enforcement services for the Immigration and Naturalization Services and other Federal, State, and local law enforcement agencies. It also develops and presents training programs in the detection of fraudulent documents, assists in identifying fraudulent documents, and acts as a liaison in promoting common efforts to combat international document fraud.

also issue birth certificates on security paper, but in 14 of the local offices with local issuance they use different security paper than the State vital records office. (One local office we visited issues birth certificates on plain white bond paper.) The security features most often used by both State and local offices are serial numbers, watermarks, and micro-line printing. Other paper security features used include intaglio and steel-engraved borders, ultraviolet ink, security threads, substrate paper or ink, hidden voids, and latent images. A chart outlining the security features contained in the paper used by State vital records registrars is located in Appendix B.

Adding to the potential number of different types of valid birth certificates in existence is the unknown number of changes in paper used to issue birth certificates, the number of different formats used, and the number of different registrars' signatures that have appeared on birth certificates since the birth registration began in most States in the early 1900's. State registrars interviewed estimate periodic changes in paper and/or formats occur every 8 to 10 years, and that registrar signatures could change more frequently, especially when registrars are local elected officials (e.g., county clerks, mayors). States and local offices also make changes in the vendors from whom they purchase their security paper. Some change vendors annually because State laws require that they purchase from the lowest bidder. Local offices also make periodic changes in the vendors from whom they order their security paper, and many order paper based on State issued specifications, but not from the same vendor used by the State.

Technological Advances Make Counterfeit Birth Certificates Easy to Create and Hard to Detect. Advances in communication (e.g., the Internet) and technologies (e.g., scanners, color printers and copiers) make it easier to obtain genuine birth certificates fraudulently or to create fraudulent birth certificates. These technological advances render the once standard methods of detecting fraudulent documents generally ineffective. With the exception of one public assistance office, which requires every applicant to be fingerprinted, virtually everyone we talked with says technology has made birth certificate fraud more likely. This is supported by the 27 primary vital records registrars responding to our survey who also say that technology has created new opportunities for fraud.

Our discussions also reveal that technology has made birth certificates the "path of least resistance" because, as other departments and agencies have increased the security features in their documents (e.g., Social Security cards, driver's licenses, and immigration documents), less secure birth certificates have been targeted for counterfeiting. As previously noted, counterfeit birth certificates are then used to obtain genuine Social Security cards, driver's licenses, and, ultimately, new identities. With each genuine document acquired, a person's new identity becomes harder to detect.

Information available through the Internet poses a significant security risk to the integrity of birth certificates. Not only are birth certificates available through the Internet, but the vital statistics necessary to request some birth certificates can be found using the Internet as well. Some vital records offices also post indexes, containing the names and birth dates of people for whom they have registered births, on their websites.

At least one local vital records office, at one time, posted scanned copies of actual birth certificates on their website. Many hospitals also have established websites where they post newborn information, and independent companies offer websites that allow parents to share their newborn's vital records information with anyone who accesses the site. We found many websites offering information to assist in creating new identities and websites offering "novelty" fake identification (many of which include birth certificates, Social Security cards, and driver's licenses). Federal and State staff we spoke with said the price "on-the-street" for a new identity, which includes a birth certificate, and often a driver's license and Social Security card, is between \$500 and \$1,200.

Most Birth Certificate Fraud is Committed Using Genuine Documents. An alarming fact is that most of the fraudulent documents identified by Immigration and Naturalization Services staff are genuine documents held by imposters, the most difficult frauds to detect. For example, staff in the El Paso Intelligence Center⁴ indicate that 90 percent of the false claims cases they see involve bonafide birth certificates held by imposters. Passport Services staff report parallel statistics, stating that 85 percent of the birth certificate fraud they encounter also is the result of genuine birth certificates held by imposters. Further, vital records registrars responding to our survey indicate they have encountered persons impersonating others to obtain genuine copies of those persons' birth certificates, individuals who had purchased birth certificates from the persons' named on the birth certificate, and persons who had stolen or acquired stolen birth certificates.

Federal and State agency staff say genuine documents held by imposters are difficult to detect. They also note that increased customer service is impacting their abilities to detect fraudulent documents and say serving a large number of customers with limited resources leaves little time to examine identification documents. We highlight our discussions with them below.

- *Public Assistance* and *Food Stamps* agency staff say they detect virtually no birth certificate fraud.
- SSA staff report detecting only limited attempts at birth certificate fraud.
- *Immigration and Naturalization Service* regional office staff indicate they may only be detecting common alternations and obvious counterfeits.
- *Passport Services* staff note fraudulent birth certificates are getting harder to detect and that imposters are almost impossible to detect once they have obtained a genuine passport.

⁴ The El Paso Intelligence Center is staffed by the Drug Enforcement Agency, Immigration and Naturalization Services, and 13 other agencies to collect information on drug trafficking, immigration violations, and other crimes.

Agencies Receive Little Training Focused on the Detection of Fraudulent Birth Certificates. Staff who depend on birth certificates as proof of citizenship and eligibility for services or benefits report receiving little training focused on the detection of fraudulent birth certificates, and that training received is focused on “obvious attempts at counterfeiting.” As such:

- SSA district and field office staff, and *Department of Motor Vehicles* office employees indicate they receive only limited training focused on the identification of fraudulent birth certificates.
- *Public Assistance* and *Food Stamps* agency staff note receiving little or no training to assist them in detecting fraudulent birth certificates.
- *Immigration and Naturalization Service* staff receive internal training to assist them in the identification of fraudulent birth certificates, but it is limited.
- *Passport Services Staff* in regional offices and the National Passport Services Center are responsible for issuing passports and receive training focused on birth certificate fraud. However, the over 4,500 acceptance agents located primarily in Post Offices and State and Federal Courts receive little training in the identification of fraudulent birth certificates.

State Practices Create Opportunities for Fraud

Delayed, Amended, and Midwife Birth Registrations Provide Opportunities for Fraud. State and local vital records staff say birth certificates issued based on delayed and amended birth registrations are more likely to be fraudulent. They also say they consider births registered by midwives, and other home births, to have a high potential for fraud.

Delayed birth registration occurs when a certificate of birth is not filed within the time specified by State law. Delayed birth registrations are sometimes the result of unattended home births, midwife births, and other out-of-hospital births. *The Model State and Vital Statistics Act and Regulations* require delayed certificates to be issued for births not filed within 1 year. Based on our survey responses, the State median time frame for filing delayed registrations is 1 year, but varies from State-to-State and ranges between 10 days and 4 years.

Federal and State staff alerted us to problems with delayed birth registration. These problems arise from the lack of evidence required to file a delayed birth in some States. Not all States include information with birth certificates about the documentary evidence they accept as proof a birth occurred, and upon which delayed registrations are allowed, with delayed birth certificates. Likewise, the documentary evidence required to register delayed births is inconsistent among States. Forty-seven States accept affidavits of

personal knowledge as proof a birth occurred, and 14 include no abstract of documentary evidence when issuing delayed birth certificates.

Amended birth registration occurs when changes are made to the vital information contained on the original birth registration. While we did not specifically address amended birth registrations in our survey, they were identified as problematic in our discussions with Federal agency and State vital records office staff. The overall concern surrounding amended registrations is similar to that of delayed registrations in that some States do not require substantial evidence to amend birth registrations and that birth certificates issued based on amended registrations are not clearly marked as having been “amended.” One State registrar also noted a growing problem in which adults are adopted by other adults, usually for inheritance purposes. In these cases, the adopted person undergoes a legal change of name and their birth record is altered, but the fact that the adoption took place is never recorded in any way on the original or amended birth certificate.

Midwife birth registrations were also identified as an area of concern. Midwives provide a valuable service in insuring the healthy delivery of children and accurate registration of births. However, out-of-hospital births attended by midwives have raised concerns. Sixteen State registrars indicate they have encountered problems specifically linked to midwife birth registration. Our discussions with Federal, State, and local staff during our onsite visits indicate that problems associated with midwife registrations are concentrated along the United States-Mexico border. In fact, midwife registration has become such a problem in one border city we visited that they now require a police officer to be called to the scene shortly after any midwife delivery to verify that the birth actually occurred in the United States. All 41 States that allow midwives to register births have procedures and guidelines in place for such registration, but only 17 of those States require information in addition to or different from that required for hospital births. The additional information required to register midwife births in these States can include attendant affidavits, prenatal and/or post-partum records, and notarized statements or other documentation verifying the birth took place. In addition, some States require that midwives provide documentation that the mother lived in the State at the time the birth occurred.

Delays in Matching Death and Birth Records Create Opportunities for Fraud.

While our contacts all agreed that matching death and birth information is a strong deterrent to improper use of a genuine birth certificate, delays in matching those records present opportunities for fraud. In our interviews, we heard many stories in which the identity of a deceased person was assumed by obtaining a copy of their birth certificate. In these stories, imposters chose their new identities from names listed in newspaper obituary columns, from newspaper articles about individuals who had been murdered or killed in accidents, or from cemetery markers. Others had assumed the identities of deceased friends or family members. In each case, the imposter obtained a copy of the deceased person’s birth certificate before it had been matched with the death registration.

With the exception of one State (Connecticut), primary vital records registrars responding to our survey say their offices match death and birth registrations and that they send information regarding deaths occurring in their State to the States in which the deceased individuals were born. They indicate most death information is transmitted to other States by mail (47), but some States (12) also transmit information electronically, and 1 State transmits death information to other States by facsimile.

Forty registrars indicate the time between when a death is registered and the time the death is recorded on a birth record creates opportunities for fraud, and 24 registrars say they do not consider the receipt of death information from other States to be timely. Thirty registrars indicate that, on the average, they provide batched death notification to other States between 1 and 3 months after the death, 11 say between 3 to 6 months after death, 4 say between 6 and 12 months, and 4 say it takes them 12 months or longer to provide death information to other States. The result is that, even though a State may send information to other States on a monthly basis, because of delays and batching of records, the deaths may have occurred many months before the death and birth records can be matched.

Once received, limited resources have forced most States to establish priorities to determine which death and birth records will be matched, and, as a result, 40 States have established priorities for matching death and birth records. As such, 20 States give priority status to matching death registrations for infants under the age of 1 year, and 17 States assign priority status to deceased individuals under the age of 45.

Adding to the problems associated with birth certificates of deceased individuals is the fact that five State registrars do not mark original birth records maintained in the State vital records office “deceased,” and six do not issue certified copies clearly marked “deceased.” In addition, birth certificates issued from computerized abstracts by those offices are not marked “deceased” in 3 of the 37 States that issue them. When asked if they would attempt to gain additional information if the information provided to them was insufficient to provide a positive match between the death registration and a birth record, only 27 registrars said yes.

Physical Security Creates Opportunities for Fraud. Despite the statements of registrars, we observed a number of questionable security situations during our onsite visits. Fifty-two of the responding registrars say they keep the paper used to print birth certificates in secured areas (e.g., locked cabinets, vaults, safes, and locked rooms) outside business hours. During business hours however, we observed a number of instances where birth certificate information or the paper used to print birth certificates were not secured. State registrars also indicate the number of vital records employees with access to birth certificate paper ranges between 3 and 60.

In our visits to local offices we also observed a number of situations we consider vulnerable. For instance, in one local office:

- certified copies of birth certificates ready for customer pick-up were kept in a box just inside the customer service counter, in plain view of the public, and well within arms-reach of anyone;
- office copies of carbon-set applications used over the last several years to request copies of birth certificates (and containing all the information needed to obtain additional copies) were kept rubber-banded together and stacked in open boxes on the floor just inside the back door; and
- several stacks of unclaimed birth certificates (some printed as far back as 1995) were kept on a shelf in a small locked closet.

Destruction of Documents - The destruction of documents and tools also poses a security problem in some offices. Some State registrars told us they are required to contract with the lowest bidder for destruction of unused or voided birth certificate paper, applications, microfilm, and embossing equipment. In some States, vital records staff “hand-shred” documents and place them in open containers for pick-up and machine shredding at a later date. In one State, paper slated for destruction was stolen or sold, and ultimately used to create fraudulent birth certificates. In that same State, applications and voided documents are now “hand-shredded” and stored in boxes awaiting a new contract for the destruction of documents.

Employee Background Checks - In spite of the fact that 25 State registrars say that birth certificate fraud has been committed by vital records employees in their State, only 14 States conduct background checks on vital records office employees.

Use by Sports Teams - Another security issue and opportunity for fraud brought to our attention was the use of birth certificates by sports team coaches to prove age and eligibility of players. Every day, thousands of certified copies of birth certificates are carried around on clip boards at ball parks across the country, and in one known instance, a coach discarded birth certificates in a trash container at the end of a season. In other instances, birth certificates used for sports activities were alleged to have been sold across the border to assist illegal aliens to gain entry into the United States.

Access to Computer Files - In our discussions with Federal and State staff, we noted a number of staff with direct access to vital records information via computer terminals. While such access allows SSA, public assistance, and Food Stamps staff to quickly verify birth record information online, it increases the number of people with access to that information. In one office, the password to enter the system was printed on a piece of paper taped to the edge of the computer screen.

Limited Oversight of Local Offices Makes Them Vulnerable to Fraud. Of the 42 States with local issuance, State vital records staff conduct monitoring visits to the local offices in their States on an annual basis in only 18 States and biannually in only 3 States. However, we question some States' abilities to conduct monitoring of between 300 and 500 local offices annually or biannually. In 10 other States, registrars say they also conduct monitoring visits to local issuing offices, but do so only on an "as needed" or occasional basis. There are 2,767 local issuing entities in States where monitoring is done "as needed" or occasionally.

State registrars say the activities conducted during onsite visits can include auditing records, reviewing laws and regulations with issuing staff, providing onsite training, and reviewing security procedures and issues. Many local offices have never been visited by State vital records staff. Staff in those offices that have been visited say the visits are usually nothing more than quick audits of their financial records related to the sale of birth certificates. In another State, the large number of local offices prohibits visiting all of them, so a self-assessment form is used to gather information about security in those offices. By completing this survey, the State vital records office not only gains information about the local offices, but it also educates local office staff regarding security measures.

Very few state vital records staff are assigned responsibility for the detection of fraud, and only 24 States employ someone responsible for fraud activities. However, in 16 States, registrars indicate staff designated responsibility for fraud activities devote less than 25 percent of their time to birth certificate fraud. Other State registrars say their lack of funds prohibits them from employing someone designated responsibility for birth certificate fraud.

In addition, Immigration and Naturalization Service and Department of Motor Vehicles office staff say they experience difficulty in obtaining the information they need to verify questionable birth certificates in a timely manner. As such, Immigration and Naturalization Service staff described situations in which they have detained individuals with suspect birth certificates for which they need immediate vital information, but were unable to get the information quickly. Passport Services staff say they also have encountered problems verifying birth certificate information in a timely manner. In addition to contacting State vital records offices directly, they contact the SSA office in Baltimore, MD, for information to assist them in determining whether or not a birth certificate presented by a passport applicant is valid. However, they are limited in the number of such requests they can make, and are forced to prioritize their requests, choosing which questionable or suspect applications they pursue.

Birth Certificate Fraud is Seldom Prosecuted

The False Identification Crime Control Act of 1982, Public Law 97-398, amended Title 18 of the United States Code to provide penalties for certain false identification related crimes.

Section 1028 of that law addresses fraud activity in connection with identification documents. This law makes it a crime to knowingly produce, transfer, or possess false identification documents with the intent to defraud the United States. Punishment for an offense under this Act is a fine of up to \$25,000 and/or imprisonment for up to 5 years. Many State statutes also address identity fraud, but it is considered a felony in only 23 States. In 28 States, birth certificate fraud remains a misdemeanor.

However, virtually all of the Federal and State staff we talked with indicate birth certificate fraud is seldom prosecuted unless it can be linked to large dollar losses or other punishable crimes. Most staff also indicate that many prosecutors are reluctant, or refuse to take birth certificate fraud cases in which the only charge is attempting to obtain another individual's birth certificate, or counterfeiting or altering a birth certificate. Further, State registrars indicate they receive information about potential fraud cases from Passport Services, SSA, Department of Motor Vehicle offices, and law enforcement. However, most indicate they receive little feedback regarding cases they refer for prosecution.

Misconceptions Exist Regarding the Security and Integrity of Birth Certificates

During the course of our study, we found that many misconceptions exist surrounding the security and integrity of birth certificates. We also found that a lack of education exists regarding the importance of securing vital records information. Misconceptions include the following:

- a birth certificate insures the identity and citizenship of the person holding it;
- birth certificates can be obtained only by the individual listed on the certificate or appropriate family members;
- if a birth certificate contains a crimped seal, it is “real”;
- birth certificate information is always safeguarded when provided to user agencies or entities (e.g., sports managers); and
- lamination improves the integrity of birth certificates.

CONCLUSIONS

Birth Certificates Alone do not Provide Conclusive or Reliable Proof of Identity

However, agencies who rely on birth certificates as a means of establishing identity must understand the limitations of accepting a birth certificate as proof of age, citizenship, or identity. While the agency has a need or requirement for establishing identity, the issuing entity most likely does not.

A related point is that genuine documents obtained with counterfeit birth certificates can be used to obtain genuine birth certificates, which in turn are used to obtain additional identification documents. This vicious cycle enables persons establishing false identities to acquire the documentation and proof of identity necessary to open bank accounts, establish credit, and evade law enforcement. Thus, it is inherently illogical to require someone to prove their identity using potentially fraudulent identity documents spawned by false birth certificates in order to obtain a birth certificate.

Further, it would be Impractical to Redesign Birth Certificates to Make them Reliable Identification Documents in and of Themselves

The birth certificate does not provide positive proof of identity of the holder. Efforts to make the birth certificate into a reliable identity document are complicated by the more than 14,000 different versions of legitimate birth certificates currently in existence, and the more than 6,000 entities that issue them and the processes used to do so. Efforts are also complicated by the ease with which birth certificates can be obtained in open States, through the mail, and via the Internet. In addition, technology has made birth certificates easier to counterfeit, and the majority of birth certificate fraud is now being committed by imposters using genuine birth certificates. Finally, because birth certificates do not expire, currently valid birth certificates will remain valid until an individual's death, meaning that changes to the birth certificate itself will take essentially a lifetime to become fully effective.

Some Efforts to Redesign Birth Certificates Might Even be Undesirable

The primary purpose for which birth certificates were created -- to document and record births -- is served well by the large number of entities that issue them and the technology that makes them readily and quickly available. It is through the many issuing entities and different issuing processes that we are afforded quick and easy access to our birth and other vital records information.

This intended purpose of birth certificates is also well served by the different types of birth certificates issued, which allows individuals to obtain information in formats most usable to them. However, these are the very things that breed fraud and render birth certificates unreliable proof of identity.

Nevertheless, Since Birth Certificates can Play an Important Role in Establishing Identity, Their Integrity Should be Improved

When used in combination with other documents, birth certificates can add to the level of proof in establishing eligibility and identity. As such, many agencies and organizations request that individuals provide birth certificates in order to receive a benefit or service, or to support the issuance of other documents used for identity purposes. The SSA uses birth certificates as proof of age in assigning Social Security numbers, and as proof of age and citizenship in determining eligibility for Supplemental Security Income benefits. Passport Services considers birth certificates primary evidence of birth in the United States, and while they do not consider birth certificates evidence of identity, they do consider them a primary document in establishing identity. In addition, public assistance agencies use birth certificates as proof of applicant age and relationship/parentage, and driver's license offices require them as proof of birth and age. Many other State and Federal agencies also rely to some extent on birth certificates as part of the process to establish personal identity and eligibility for services and benefits.

Given that birth certificates will undoubtedly continue to be an important element in determining identity, it is important that the processes used to issue birth certificates be standardized and that recent advances in technology be utilized to ensure birth certificate integrity parallels that of other identification documents. In support of this conclusion, we provide the consensus responses we received for improving the birth certificate process and the detection of birth certificate fraud. The suggestions include the following:

State vital records offices and other entities that issue birth certificates may wish to consider

- taking the steps necessary to reduce the number of legitimate birth certificates by substantially reducing the number of entities that issue birth certificates and different types of birth certificates issued;
- establishing national requirements for security paper;
- placing a higher priority on matching death and birth records and the speed at which these records are matched;
- reducing opportunities for fraud created as the result of delayed, amended, or midwife birth registrations by placing greater emphasis on the scrutiny of

supporting documentation allowed as verification to register delayed, amended, or midwife birth and marking delayed and amended birth registrations accordingly;

- expanding the number of staff assigned responsibility for the detection and enforcement of birth certificate fraud;
- introducing the use of biometrics (e.g., fingerprints or other individual physical identifier) into the birth certificate process, thus insuring positive links between birth certificates and the people presenting them as proof of identity; and
- launching a national campaign to inform the general public and user agencies about the importance of safeguarding vital records and their vulnerability to fraud.

In Addition, Federal and State Program Administrators Should Assess the Proofs of Identity They Will Accept

Even if their security is improved, birth certificates may still not be the best proof of identity. For this reason, program administrators may not want to use birth certificates at all, or to use them only with other documents, as noted above. Agencies need to specify which other documents and methods of proving identity (e.g., fingerprints, testimony of relatives, employer information, background checks) they will accept in determining eligibility for services and benefits.

Given what we have learned, if program administrators continue to include birth certificates in the proofs of identity their agencies and organizations will accept, they should reconsider how they use them, and what steps they will take to detect fraudulent certificates and to secure valid ones. For example, they should remain vigilant in their detection of fraudulent documents and genuine documents held by imposters. The nature of each individual program should dictate the level of proof the agency requires to establish identity, and acceptable documents determined accordingly. For instance, the same level of proof required to enroll a child in school may not be the same level of proof required to obtain a passport.

In this vein, we provide consensus responses from persons surveyed and interviewed, which reflect longstanding suggestions for improving the use of birth certificates in establishing identity.

Federal and State agencies who use birth certificates in determining eligibility for services and benefits may wish to consider

- reassessing the documents they will accept as proof of identity and program eligibility;
- accepting only birth certificates issued on security paper that meets current national standards;

- using biometrics (e.g., fingerprints, DNA, retinal scans) to assist the agencies in establishing proof of identity;
- improving guidelines and procedures regarding the detection of fraudulent birth certificates, and make them readily accessible; and
- providing staff with ongoing training to assist them in detecting all types of birth certificate fraud.

State and Local Birth Certificate Issuing Entities

APPENDIX A

State	Number of Issuing Offices by Location					
	Central/State	County	City	Township	Other*	Total
Alaska	3	0	4	15	0	22
Alabama	1	67	0	0	1	69
Arkansas	1	0	0	0	0	1
Arizona	1	12	0	0	2	15
California	1	58	4	0	0	63
Colorado	1	63	0	0	0	64
Connecticut	1	0	169	0	0	170
Washington, DC	1	0	0	0	0	1
Delaware	1	2	0	0	0	3
Florida	1	67	0	0	0	68
Georgia	1	159	0	0	0	160
Hawaii	1	0	0	0	0	1
Iowa	1	100	0	0	0	101
Idaho	1	0	0	0	0	1
Illinois	1	42	100	2	0	145
Indiana	1	91	3	0	0	95
Kansas	1	0	0	0	0	1
Kentucky	1	0	0	0	0	1
Louisiana	1	0	0	0	5	6
Massachusetts	1	0	351	0	0	352
Maryland	1	20	0	0	1	22
Maine	1	0	22	475	0	498
Michigan	1	83	27	0	0	111
Minnesota	1	87	5	0	5	98
Missouri	1	113	3	0	0	117
Mississippi	1	0	0	0	0	1
Montana	1	56	0	0	0	57
North Carolina	1	98	0	0	3	102
North Dakota	1	0	0	0	0	1
Nebraska	1	1	0	0	0	2
New Hampshire	1	0	13	222	0	236
New Jersey	1	0	565	0	0	566
New Mexico	1	9	0	0	0	10
Nevada	1	2	0	0	0	3
New York	1	4	100	1,000	400	1,505
Ohio	1	88	55	0	0	144
Oklahoma	1	1	1	0	0	3
Oregon	1	35	0	0	0	36
Pennsylvania	1	0	0	0	5	6
Puerto Rico	1	0	0	84	0	85
Rhode Island	1	0	39	0	0	40
South Carolina	1	46	0	0	0	47
South Dakota	1	64	0	0	0	65
Tennessee	1	23	0	0	4	28
Texas	1	278	587	0	0	866
Utah	1	0	0	0	12	13
Virginia	1	0	0	0	0	1
Vermont	2	0	8	238	0	248
Washington	1	33	0	0	0	34
Wisconsin	1	72	2	0	0	75
West Virginia	1	55	0	0	0	56
Wyoming	1	0	0	0	0	1
NY City	1	0	0	0	0	1
Total	560	1,829	2,058	2,036	438	6,417

*Includes villages, regional branch offices, and some State vital records satellite office.

Types of Certificates Issued and Paper Security Features Used by State Vital Records Offices

	Type of Birth Certificate			
	Full Photocopy of Birth Record	Certified Copy from Computerized Abstract	Birth Card or Wallet Sized Birth Certificate	Commemorative Birth Certificate
Number of States Issuing Type of Birth Certificate	51	37	17	8

Number of States Using Specified Security Features

Watermarks	27	25	6	3
Intaglio	19	20	9	2
Serial Numbers	35	34	15	7
Steel Engraved Borders	18	19	10	2
Ultraviolet Ink	9	13	2	1
Security Threads	11	13	3	0
Micro-Line Printing	25	28	7	1
Substrate Paper or Ink	14	14	4	1



Vital Records and Vital Statistics in the United States: Uses, Users, Systems, and Sources of Revenue

Conducted for the National Committee on Vital and Health Statistics
Subcommittee on Population Health

January 10, 2018

This report was prepared under contract by R. Gibson Parrish, M.D.

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Background and Purpose

The National Committee on Vital and Health Statistics (NCVHS) serves as the advisory committee to the Secretary of Health and Human Services on health data, statistics, privacy, standards, national health information policy, and the Health Insurance Portability and Accountability Act (HIPAA). In this capacity, it provides advice and assistance to the Department and serves as a forum for interaction with stakeholders on key issues related to population health, standards, privacy and confidentiality, and data access and use. Because the national vital statistics system (NVSS) plays a critical role in monitoring the health of the nation, conducting public health surveillance, understanding the effectiveness of the healthcare and health financing system, and supporting business and commerce, the NCVHS is exploring the state of the existing NVSS and identifying ways to improve the system. The goal of NCVHS' exploration is the answer to the following question:

“How do we transform today’s vulnerable vital records data collection network into a network of state systems that produce accurate and timely information supporting a breadth of local, state and federal data needs?”

As part of this exploration, NCVHS held a “Hearing on Next Generation Vital Statistics” in Washington, D.C., on September 11–12, 2017. The hearing’s objectives were to:

- (1) identify the essential elements of the vital statistics system;
- (2) assess the system’s current status and risks to its viability; and
- (3) consider what actions are needed to protect and improve the system.

At the hearing, more than 25 agencies, organizations, and businesses testified about their use of, and reliance on, vital records and statistics for carrying out their day-to-day activities. Representatives of the NVSS at the Federal, state, and local level also testified about the current status of the NVSS and its ongoing challenges. The agenda, testimony, and a transcript of the hearing are available on the NCVHS website,¹ and a report summarizing the hearing is being prepared.

Following the hearing, members of NCVHS with the support of the National Center for Health Statistics (NCHS) determined that it would be useful to develop a companion report that would provide a summary of the uses and users of vital records and vital statistics and an analysis of the types and sources of revenue available to vital records jurisdictions in the United States

The purpose of this report is to present the uses, users, and sources of revenue for vital records and vital statistics. The information provided in this report contributes to the input being considered by the Committee as it identifies and considers recommendations for approaches that would address the many challenges facing the NVSS at this time.

¹ NCVHS. Agenda: Next Generation Vital Statistics: A Hearing on Current Status, Issues and Future Possibilities, Hubert H. Humphrey Building, Room 705A, Washington, DC; September 11-12, 2017. Available at <https://www.ncvhs.hhs.gov/meeting-calendar/agenda-of-the-september-11-12-2017-subcommittee-on-population-health-next-generation-vital-statistics-hearing/>

Terminology

Several terms are used to describe different aspects of the NVSS:

- (1) Vital events are the major life events for individuals and include fetal death, still birth, birth, marriage, divorce, and death. The focus of this report is on two vital events: birth and death.
- (2) Vital registration is the process of collecting, recording, and storing specific information on individual vital events. When this process is completed electronically, it is referred to as “electronic [vital event] registration”; examples include electronic birth registration and electronic death registration.
- (3) Vital records are the product of vital registration and consist of the completed paper or electronic forms that document vital events, such as birth certificates and death certificates. Vital records may be stored in either a paper or an electronic form.
- (4) Vital registration system consists of the various persons, agencies, equipment, forms, rules (standards), and processes that carry out vital registration. When this system completes vital registration electronically, it is referred to as an “electronic [vital event] registration system”; examples include electronic birth registration system (EBRS) and electronic death registration system (EDRS).
- (5) Vital records office (VRO) is the government agency or organization within a state, local, or territorial jurisdiction that is responsible for registering vital events, storing vital records, and issuing certified copies of vital records for legal and administrative purposes. In some jurisdictions, the VRO may be referred to as the “vital registration office” or the “vital statistics office.”
- (6) Vital statistics consists of quantitative data concerning vital events in a population, such as the number of births and the death rate. Vital statistics are derived from an analysis of vital records data.

Approach

NCVHS contracted with R. Gibson Parrish, M.D., to prepare this report under the guidance of the NCVHS Subcommittee for Population Health, and Kate Brett, Lead Staff to the Population Health Subcommittee, and Rebecca Hines, NCVHS Executive Secretary, both with NCHS. Dr. Parrish prepared the report during October and November 2017.

To identify uses and users of vital records and vital statistics and the types and sources of revenue available to vital records offices in the United States, Dr. Parrish reviewed NCVHS Hearing-related materials, including the agenda, submitted testimony, and transcripts. Based on this review, he searched the Internet for agencies and organizations that produce, manage, support, or use vital records or vital statistics. The web sites identified in these searches are listed in association with the relevant table or tables in the findings section of this report and in this report’s list of references. Dr. Parrish also interviewed or corresponded with the following individuals to gain additional information on the NVSS, and he appreciates the assistance that they provided:

- (1) Delton Atkinson, Director of the Division of Vital Statistics at the National Center for Health Statistics, Centers for Disease Control and Prevention
- (2) Kate Brett, Epidemiologist in the Division of Vital Statistics at the National Center for Health Statistics, Centers for Disease Control and Prevention
- (3) Bruce Cohen, National Committee on Vital and Health Statistics
- (4) Rebecca Hines, Health Scientist, National Center for Health Statistics, Centers for Disease Control and Prevention

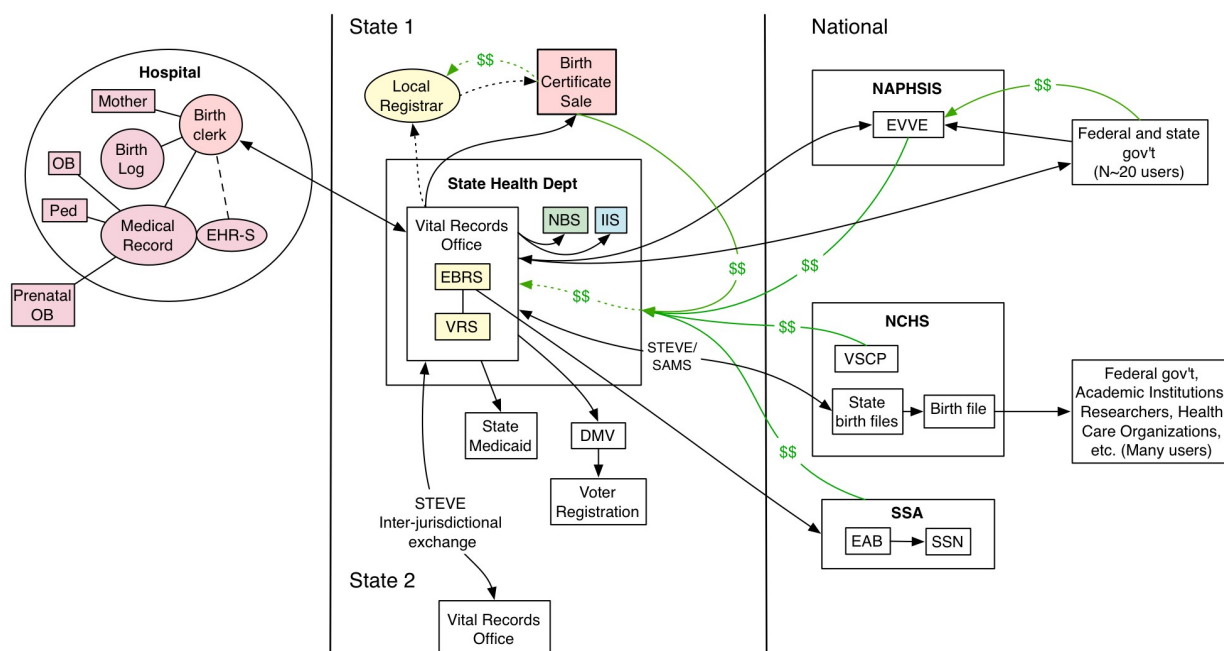
- (5) Rich McCoy, Director of the Center for Health Statistics at the Vermont Department of Health, and President-Elect of the National Association of Public Health Statistics and Information Systems
- (6) Andrea Price, National Association of Public Health Statistics and Information Systems
- (7) Steven Schwartz, Registrar of the Bureau of Vital Statistics at the New York City Department of Health and Mental Hygiene
- (8) Anthony Stout, National Association of Public Health Statistics and Information Systems
- (9) Shawna Webster, Executive Director, National Association of Public Health Statistics and Information Systems

Findings

Overview of birth and death registration

Figures 1 and 2 provide overviews of the birth and death registration processes, respectively. Both figures show the process of gathering information about the vital event on the left, including the types of people and organizations involved in the process. The middle panels depict the roles of state government in registering the vital event and providing data about vital events to various state and national users. The right panels depict major national agencies and organizations that compile, use, and distribute birth and death records obtained from state vital records offices. The figures also depict payments made by various users of vital records to state or national organizations for the use of the records.²

Figure 1. Current birth registration process for a state vital records jurisdiction



Legend for Figure 1:

— Solid black and green lines denote typical processes and payments, respectively

- - - Dashed black and green lines denote other, less common processes and payments, respectively

² Payments by users to providers of vital records or vital statistics data are shown by green lines with “\$”.

DMV = Department of Motor Vehicles

EAB = Enumeration at Birth

EBRS = Electronic Birth Registration System

EHR-S = Electronic Health Record System

EVVE = Electronic Verification of Vital Events

IIS = Immunization Information System

NBS = Newborn Screening

NCHS = National Center for Health Statistics

OB = Obstetrician or comparable health care provider

Ped = Pediatrician

SAMS = Secure Access Management Services

SHD = State Health Department

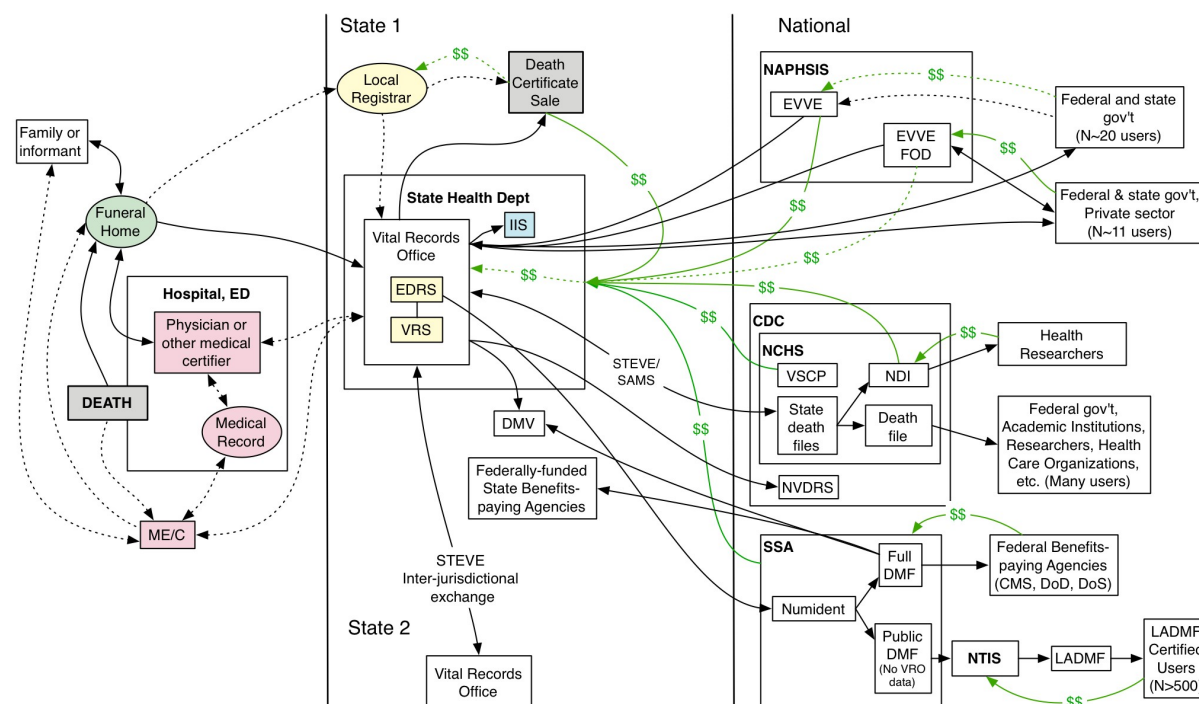
SSA = Social Security Administration

SSN = Social Security Number

STEVE = State and Territorial Exchange of Vital Events System

VRS = Vital Registration System

VSCP = Vital Statistics Cooperative Program

Figure 2. Current death registration process for a state vital records jurisdiction


Legend for Figure 2:

— Solid black and green lines denote typical processes and payments, respectively

--- Dashed black and green lines denote other, less common processes and payments, respectively

CMS = Centers for Medicare and Medicaid Services
 CR = Cancer Registry
 DMF = Death Master File
 DMV = Department of Motor Vehicles
 DoD = U.S. Department of Defense
 DoS = U.S. Department of State
 ED = Emergency Department
 EDRS = Electronic Death Registration System
 EHR-S = Electronic Health Record System
 EVVE = Electronic Verification of Vital Events
 EVVE FOD = Electronic Verification of Vital Events Fact of Death
 IIS = Immunization Information System
 LADMF = Limited Access Death Master File

ME/C = Medical Examiner / Coroner
 NCHS = National Center for Health Statistics
 NDI = National Death Index
 NTIS = National Technical Information Service
 NVDRS = National Violent Death Reporting System
 SAMS = Secure Access Management Services
 SHD = State Health Department
 SSA = Social Security Administration
 STEVE = State and Territorial Exchange of Vital Events System
 VRO = Vital Records Office
 VRS = Vital Registration System
 VSCP = Vital Statistics Cooperative Program

Uses and users of identified birth and death records

Uses and users of identified³ birth and death records are shown in Tables 1 and 2, respectively. These uses and users include individuals who use their birth certificate to establish their identity for various legal and administrative purposes; the Social Security Administration (SSA), which uses birth data to facilitate the issuance of Social Security Numbers (SSN) to persons born in the United States and death data to terminate the payment of benefits; and private organizations, which use death data to pay benefits and to prevent fraud.

³ "Identified records" refers to one or more of records, each of which contains information sufficient to identify the subject of the record, such as name, name of mother, and location and date of birth on a birth record.

These identified birth and death records are available through state and other vital records offices (VROs) to authorized individuals, who usually must pay a fee for copies or use of the records. Verification of birth and death records is also available to authorized users for all state and other vital records jurisdictions—except New York and Texas—through the Electronic Verification of Vital Events (EVVE) system, which is maintained by the National Association of Public Health Statistics and Information Systems (NAPHSIS). NAPHSIS also operates the Electronic Verification of Vital Events Fact of Death (EVVE FOD) system, which provides authorized users with the ability to discover if a death record exists for an individual. Currently, 42 states and jurisdictions participate in the EVVE FOD system. (See Appendix C and Table 5.)

The Social Security Administration purchases identified birth and death records from VROs to administer its benefits programs. These records form a part of the NUMIDENT electronic database, which contains SSA's records of Social Security Numbers (SSN) assigned to individuals since 1936. The SSA compiles its death information in the Death Master File (DMF), which is an extract of death information on NUMIDENT (see Appendix C). The DMF includes, if available, each deceased individual's SSN, first name, middle name, surname, date of birth, and date of death. The SSA provides access to the "full" DMF for a "reasonable cost" to Federal or state agencies that provide federally funded benefits to individuals. The SSA may also provide State driver's license agencies with access to its records to verify applicable information "pursuant to the Help America Vote Act of 2002." (See Appendix C and Table 5.)

A "public" version of the DMF contains death records extracted from the NUMIDENT database, but does not include death data received from the states. The "public" version of the DMF is available from the Department of Commerce's National Technical Information Service (NTIS) as the "Limited Access DMF," or "LADMF," to certified persons and organizations, who must have a legitimate fraud prevention interest, or have a legitimate business purpose pursuant to a law, governmental rule, regulation, or fiduciary duty. The NTIS charge fees for using the LADMF and to cover the costs associated with its certification program. (See Appendix C and Table 5.)

NCHS established and maintains the National Death Index (NDI), a centralized database of death record information on file in state vital statistics offices, as a resource to aid epidemiologists and other health and medical investigators with their mortality ascertainment activities. Researchers pay a fee for using the NDI and can request either a *routine* NDI search for those who do *not* need cause of death codes, or an NDI *Plus* search, which also provides cause of death codes. (See Appendix C and Table 5.)

Table 1. Uses of identified birth records and data sets

Use	User(s)	System	Records used	VRO Funding source
<i>Individual or Family</i>				
Establish identity for legal and administrative purposes	Individual or family	State and jurisdiction VROs	1	Sale of certificate
Obtain or renew U.S. Passport	Individual or family	State and jurisdiction VROs	1	Sale of certificate
Provide proof of citizenship or legal residency for employment	Individual person	State and jurisdiction VROs	1	Sale of certificate

Verify identity for drivers license or identity card	Individual person	State and jurisdiction VROs	1	Sale of certificate
Verify age of child for sports league	Individual or family	State and jurisdiction VROs	1	Sale of certificate
Provide proof of country of birth for college admission	Individual person	State and jurisdiction VROs	1	Sale of certificate
Federal Government				
Verify and certify birth information	SSA	EVVE*	1 to many	User fee
	DoS/Diplomatic Security	EVVE*	1 to many	User fee
	OPM	EVVE*	1 to many	User fee
	Regional FBI Offices	EVVE*	1 to many	User fee
	DHS/USCIS	EVVE*	1 to many	User fee
Issue Social Security Number (SSN) to each newborn to prevent other parties from filing a request for a SSN in the name of the newborn	SSA	State and jurisdiction EBRS	All births	SSA contracts with State VROs
Obtain or renew U.S. Passport	DoS/Passport Services Fraud Prevention Offices	State and jurisdiction VROs	1 to many	Unknown†
		EVVE*	1 to many	User fee
Conduct evaluations of state Medicaid programs	CMS	State and jurisdiction VROs	1 to many	CMS contractor payments to VRO
Obtain birth certificate demographic and medical data for the National Vital Statistics System	NCHS	State and jurisdiction EBRS	All births	NCHS (VSCP)
State and Local Government				
Assist state and local government agencies in performing their official duties*	Secretary of State	EVVE*	Selected births	User fee
Enroll eligible children into Medicaid program	State Medicaid programs	State and jurisdiction VRO/EBRS	Selected births	Unknown†
		EVVE*		User fee
Verify identity for driver's license or identity card	DMV	State and jurisdiction VROs	Selected births	Unknown†
		EVVE*		User fee
Verify birth for voter	DMV	SSA birth file	Selected	None: Birth

registration			states and births	data obtained from SSA
Initialize entry in Immunization Information System (IIS)	State IIS	State and jurisdiction EBRS	All	Unknown†
Facilitate newborn screening for health disorders	Health care provider responsible for screening newborn	State and jurisdiction VRO/EBRS	All	Unknown†
Private (Non-Government) Organizations				
Genealogical research	Individual person or organization	State and jurisdiction VROs	1 to many	Sale of certificate(s) or VR data
Verify age of child for sports league	Sport league	State and jurisdiction VROs	1 to many	Sale of certificate(s)
Use by media to verify identity or other information	Media organizations	State and jurisdiction VROs	1 to many	Sale of certificate(s) or VR data
Use of birth data for medical or other health research	Medical and other health researchers	State and jurisdiction VROs	1 to many	Sale of certificate(s) or VR data
Conduct background checks for government agencies, pension funds, or other organizations (Note: Some states can't provide certificates to these third parties.)	Company under contract to government agency or pension fund (e.g., Berwyn Group)	State and jurisdiction VROs	1 to many	Sale of certificate(s) or VR data
Determine property rights and place them in land records	Title search companies	State and jurisdiction VROs	1 to many	Sale of certificate(s) or VR data
Marketing products to parents of newborns. (Note: Some states can refuse these requests.)	Marketing companies	State and jurisdiction VROs	Many	Sale of certificate(s) or VR data

* For information on births occurring in states or jurisdictions not participating in EVVE, the user must contact the state or jurisdiction's VRO for information about birth records.

† The author of this report was unable to find the information needed to fill in this cell in this table. The VRO source of funding may differ by jurisdiction.

Sources of information for Table 1:

- (1) *NAPHSIS: About EVVE FOD. Available at <https://www.naphsis.org/evve-fod>; accessed on 4 Oct 2017.*
- (2) *Personal communication with Rich McCoy, Vermont State Registrar and Chief, Center for Health Statistics, 13 Oct 2017.*
- (3) *Presentations at NCVHS hearing, September 11-12, 2017, Washington, D.C., available at <https://www.ncvhs.hhs.gov/meeting-calendar/agenda-of-the-september-11-12-2017-subcommittee-on-population-health-next-generation-vital-statistics-hearing/>; accessed on 4 Oct 2017.*

2017.

Table 2. Uses of identified death records and data sets

Use	User	System	Records used	VRO Funding source
Individual or Family				
Establish fact of death for legal and administrative purposes. Note: Death Certificate cannot be provided to funeral directors in all states.	Family of decedent	State and jurisdiction VROs	1	Purchase of certificate(s) by individual, family, or funeral director
Federal Government				
Assist federal government agencies in performing their official duties (Federal Administrative Use)*	<i>Examples of users:</i>			
	DHS	EVVE FOD†	1 to many	User fee
	DoS	EVVE FOD†	1 to many	User fee
	IRS	EVVE FOD†	1 to many	User fee
	NIOSH	EVVE FOD†	1 to many	User fee
	U.S. Treasury	EVVE FOD†	1 to many	User fee
Establish fact and cause of death to pay death benefit, or terminate payments under retirement or other benefit programs	VA	EVVE FOD†	1 to many	User fee
	DoD/Defense Manpower Data Center	Full DMF	Active duty military deaths	None: DMF obtained from SSA
	VA	Full DMF†	Veteran deaths	
	Federal Retirement Thrift Investment Board	Full DMF†	Deaths of federal civilian and uniformed services employees	
	OPM	Full DMF†	Deaths of federal civilian and uniformed services employees	
	HUD	Full DMF†	Unknown††	
Establish fact of death to terminate, redirect, or otherwise modify Social Security payments or other benefits	USDA/FSA	Full DMF†	Unknown††	
	USDA/NRCS	Full DMF†	Unknown††	
	USDA/RMA	Full DMF†	Unknown††	
Establish fact of death to terminate, redirect, or otherwise modify Social Security payments or other benefits	SSA	State and jurisdiction EDRS	All deaths	SSA contracts with State VROs
Assist federal government agencies in paying death benefits, or terminating payments under disability, unemployment, workers' compensation, pension, retirement, or other benefit programs	DoL/OWCP	Unknown††	Deaths of certain civilian employees	Unknown††

Use	User	System	Records used	VRO Funding source
Establish fact of death to terminate benefits	CMS	Full DMF†	All deaths	None: DMF obtained from SSA
	U.S. Railroad Retirement Board	Full DMF†	Deaths of railroad workers and their families	
	Pension Benefit Guaranty Corporation	Full DMF†	Deaths of persons covered by PBGC	
Verify deaths of individuals to prevent tax fraud by use of identities of recently deceased individuals to claim tax refunds	IRS	Full DMF†	All deaths	None: DMF obtained from SSA
Identify and obtain information about deaths due to specific causes	CPSC	State and jurisdiction VROs	Selected deaths	Sale of certificates
Identify health study subjects who have died	CDC	NDI	All deaths	NDI user fees
	Census Bureau			
	CMS			
	DoD			
Identify health study subjects who have died, if researcher can't obtain records quickly enough through NDI, or research proposal is rejected by NDI, or researchers only need records from one or two states	NIH	State and jurisdiction VROs	Selected deaths	Sale of certificates
	NIH or other federal agency			
State and Local Government				
Assist state and local government agencies in performing their official duties (State/local Administrative Use)*	Examples of users:		1 to many	User fees paid by state and local government agencies
	County government	EVVE FOD†		
	DMV			
	Law enforcement			
	Secretary of State			
Assist state and local government agencies in paying death benefits, or	Social services	LADMF†	1 to many	None: NTIS User fees
	See Appendix D			
Assist state and local government agencies in paying death benefits, or	State and local government pension plans	EVVE FOD†	All	User fees paid by state and local

Use	User	System	Records used	VRO Funding source
terminating payments under disability, unemployment, workers' compensation, pension, retirement, or other benefit programs (State/local Benefit Use)	Unemployment programs Workers' compensation program			government agencies
Private (Non-Government) Organizations			EVVE FOD	
Assist financial institutions in verifying legitimacy and legal ownership of financial transactions, such as investments, loans and deposits.	<i>Examples of users:</i> Citigroup Equifax Fidelity Northern Trust	EVVE FOD†	1 to many	User fees paid by financial institutions
Assist healthcare organizations in determining whether patients survive care and in removing decedents from their files (Healthcare Administrative Use)	<i>Examples of users:</i> Ascension Health Tenet Healthcare Trinity Health	EVVE FOD†	1 to many	User fees paid by health care organizations
Determine property rights and enter them in land records	Title search companies	State and jurisdiction VROs	1 to many	Sale of certificate(s)
Establish fact and cause of death to pay benefit	Life insurance companies		1	None: Certificate obtained from decedent's family
Assist insurance companies in identifying insured individuals who have died, providing warranted benefits to beneficiaries, and in complying with applicable state laws	<i>Examples of users:</i> MetLife State Farm See Appendix D	EVVE FOD† LADMF†	1 to many 1 to many	User fee None: User fees paid to NTIS
Establish fact of death to terminate payments or other benefits	Employer-based pension or retirement plans	EVVE FOD† LADMF†	Selected deaths 1 to many	User fee None: User fees paid to NTIS
Obtain data on deaths for government agencies, pension funds, or other organizations (Note: Some states can't provide certificates to these third	Company under contract to government agency or pension fund (e.g., Berwyn Group)	State and jurisdiction VROs	1 to many	Sale of certificate(s) or VR data

Use	User	System	Records used	VRO Funding source
parties.)				
Identify health study subjects who have died	Academic institutions Medical centers Private companies	NDI	Selected deaths	NDI user fees
Identify health study subjects who have died, if researcher can't obtain records quickly enough through NDI, research proposal is rejected by NDI, or researchers only need records from one or two states	Health Researchers, Academic institutions	State and jurisdiction VROs	Selected deaths	Sale of certificate(s) or VR data
Use by media to verify death or find other information	Media organizations	State and jurisdiction VROs	1 to many	Sale of certificate(s) or VR data
Genealogical research	Individual person or organization	State and jurisdiction VROs	1 to many	Sale of certificate(s) or VR data

* Government agency administrative use excludes medical or health research uses, where such research is defined as a systematic study to gain information and understanding with the goal of finding ways to improve human health and/or is designed to develop or contribute to generalizable scientific knowledge. The National Death Index (NDI)—operated by the CDC's National Center for Health Statistics—provides fact of death information for medical and health research use. (See (1) NAPHSIS: About EVVE FOD, available at: <https://www.naphsis.org/evve-fod>; accessed 4 Oct 2017) and (2) CDC: NCHS: National Death Index, <https://www.cdc.gov/nchs/ndi/index.htm>; accessed on 9 Oct 2017.

† For information on births occurring in states or jurisdictions not participating in Full DMF, EVVE, EVVE FOD, or LADMF, the user must contact the state or jurisdiction's VRO for information about birth records.

†† The author of this report was unable to find the information needed to fill in this cell in this table.

Sources of information for Table 2:

- (1) Email with Subject: Users of the 205(r) data: DMF Customers and Death Exchanges, from Tom Klouda, U.S. Senate, August 25, 2017 at 2:19:10 PM EDT.
- (2) NAPHSIS: About EVVE. Available at <https://www.naphsis.org/evve>; accessed on 4 Oct 2017.
- (3) National Technical Information Service: Limited Access Death Master File: List of persons certified to receive the Limited Access Death Master File, available at <https://classic.ntis.gov/assets/pdf/DMFcertifiedList.docx>; accessed on 9 Oct 2017.
- (4) Personal communication with Rich McCoy, Vermont State Registrar and Chief, Center for Health Statistics, 13 Oct 2017.
- (5) Presentations at NCVHS hearing, September 11-12, 2017, Washington, D.C., available at <https://www.ncvhs.hhs.gov/meeting-calendar/agenda-of-the-september-11-12-2017-subcommittee-on-population-health-next-generation-vital-statistics-hearing/>; accessed on 4 Oct 2017.

Uses and users of de-identified natality (birth) and mortality (death) data sets

NCHS compiles the birth and death records from the 57 U.S. vital records jurisdictions into several de-identified vital statistics data sets, which are made available to various users, including researchers, policy makers, government agencies, academic institutions, private organizations, and the public (see Tables 3–5). NCHS disseminates public-use data files through interactive tables, online querying tools, and direct download through its FTP file server. The NCHS Research Data Center (RDC) allows access to restricted data and offers several modes of access on-site at the NCHS facility in Hyattsville, Maryland.⁴

Over the years, Federal confidentiality standards have changed for the public release of geographic and date data on national vital statistics data sets, which is reflected in the data items available in successive time periods. National birth, death, and fetal death public-use micro-data files prior to 1989 contain the county and exact dates (year, month, and day) of birth or death. For data years 1989 to 2004, micro-data files contain only geographic identifiers for counties and cities with a population of 100,000 or greater, and no exact dates. For birth, death, and fetal death files, the year, month, and day of week (e.g. Monday) are available. Beginning with the 2005 data year these data files contain individual-level vital event data at the national level only with no geographic identifiers. These files generally include most other items and detail from the vital record with the exception of exact dates.

Subject to certain limitations, data users may also access NCHS natality and mortality data using Internet programs to construct their own tabulations of births and deaths with geographic detail. These interactive systems allow users to build tables based on micro-data; however, only tabulated data are presented to the user. Interactive Web-based systems currently available are the CDC's WONDER (Wide-ranging ON-line Data for Epidemiological Research) and WISQARS (Web-based Injury Statistics Query and Reporting System).⁵

Customized NCHS natality and mortality data files are available under restricted conditions. Researchers may request customized files containing geographic details for all states and counties. These requests require a review by NAPHSIS. If the project proposal is denied because of issues related to data confidentiality, alternative access may be available through the NCHS RDC, which has specific procedures for controlled access to micro-data files. Researchers pay NCHS for the use of the RDC to support its maintenance.

State health statistics offices also may provide access to natality and mortality data files for their state for community health assessments, government research, private studies, and public inquiries. States provide differing levels of identifying details for these files. Some states also maintain Web-based data query systems, which provide interactive access to natality, mortality, and other state data.⁶

⁴ For more information, see (1) CDC: National Center for Health Statistics. Accessing Data from the National Center for Health Statistics, available at https://www.cdc.gov/nchs/data_access/index.htm; accessed 6 Nov 2017. (2) CDC: National Center for Health Statistics. Public-Use Data Files and Documentation, available at https://www.cdc.gov/nchs/data_access/ftp_data.htm; accessed 6 Nov 2017. (3) CDC: National Center for Health Statistics. Resources for Researchers, available at https://www.cdc.gov/nchs/nchs_for_you/researchers.htm; accessed 6 Nov 2017.

⁵ WONDER is available at <https://wonder.cdc.gov>, and WISQARS is available at <https://www.cdc.gov/injury/wisqars/index.html>.

⁶ Information on uses and users of de-identified natality (birth) and mortality (death) data sets was provided by Kate Brett and Delton Atkinson, NCHS, 31 October 2017.

Table 3. Uses of de-identified natality (birth) data sets

Use	User	VRO Funding source
Prepare natality statistics for the United States	NCHS	NCHS (VSCP)
Prepare natality statistics for state and local jurisdictions	State and local offices of health statistics	State or local budget
Develop health policy	- Federal, state, and local government agencies - Academic institutions - Non-governmental, health organizations	None
Carry out health research	- Federal, state, and local government agencies - Academic institutions - Non-governmental, health organizations	None
Conduct actuarial analyses for program planning and management	- SSA - CMS	None
Prepare population estimates	Census Bureau	None

Source of information for Table 3: Kate Brett and Delton Atkinson, NCHS, 31 Oct 2017.

Table 4. Uses of de-identified mortality data sets (death and linked birth-infant death)

Use	User	VRO Funding source
Prepare mortality statistics for the United States	NCHS	NCHS (VSCP)
Prepare mortality statistics for state and local jurisdictions	State and local offices of health statistics	State or local budget
Develop health policy	- Federal, state, and local government agencies - Academic institutions - Non-governmental, health organizations	None
Carry out health research	- Federal, state, and local government agencies - Academic institutions - Non-governmental, health organizations	None
Conduct actuarial analyses for insurance, pension, and other benefit programs	- SSA - CMS - Insurance Companies - Private and Government Pension Funds	None
Prepare population estimates	Census Bureau	None

Source of information for Table 4: Kate Brett and Delton Atkinson, NCHS, 31 Oct 2017.

Table 5. Selected data sets and information systems based on vital records that provide data about vital events

System*	Responsible agency or organization	Description	Type of vital records data to which access is provided	Authorized users
Birth Data				
Birth file	CDC, NCHS	Provides record level birth data derived from state and jurisdiction birth certificates submitted to NCHS through the Vital Statistics Cooperative Program (VSCP)	Identified records [†] with demographic, geographic, and medical information for births ^{††}	Researchers who submit and receive approval for a research proposal outlining the need for “sensitive” records
			De-identified records for statistical reporting and analysis	Members of the public who are willing to agree to not use data to identify any individuals or to link data with individually identifiable data from other NCHS or non-NCHS datasets.
CDC WONDER	CDC	Provides users with the ability to quickly query and analyze birth and death data sets from the National Vital Statistics System	De-identified demographic, geographic, and medical information for births ^{††}	Members of the public who are willing to agree to not use data to identify any individuals
Web-based data query systems (WDQS)	State or local governments who maintain these systems	Provide users with the ability to query and conduct statistical analyses of birth, death, and other state or local data	De-identified demographic, geographic location, and medical information about births and deaths for analysis	Members of the public. Some WDQS may ask users to agree to not use data to identify any individuals.
EVVE	NAPHSIS	Provides users with the ability to quickly, reliably, and securely verify birth or death data with or without a paper certificate	None is provided to user. The system simply confirms or denies that there is a match of the individual in the request with a birth or death record.	State and federal government agencies
Death Data				
Multiple Cause of Death Data File	CDC, NCHS	Provides record level death data derived from state and jurisdiction birth certificates submitted to NCHS through the Vital	Identified records [†] with demographic, geographic, place, and complete cause information	Researchers who submit and receive approval for a research proposal outlining the need for “sensitive” records

System*	Responsible agency or organization	Description	Type of vital records data to which access is provided	Authorized users
		Statistics Cooperative Program (VSCP)	for death De-identified records for statistical reporting and analysis	Members of the public who are willing to agree to not use data to identify any individuals or to link data with individually identifiable data from other NCHS or non-NCHS datasets.
Linked birth-infant death file: period data	CDC, NCHS	Provides data on all infant deaths occurring in a given year (e.g., 2015) linked to their corresponding birth certificates, whether the birth occurred in the given year (e.g., 2015) or the previous year (e.g., 2014). The denominator file for the data set is the given year's natality file, that is, all births occurring in the given year (e.g., 2015).	Identified records† with demographic, geographic, and medical information about the birth and death of infants who die De-identified records for statistical reporting and analysis	Researchers who submit and receive approval for a research proposal outlining the need for "sensitive" records Members of the public who are willing to agree to not use data to identify any individuals or to link data with individually identifiable data from other NCHS or non-NCHS datasets.
CDC WONDER	CDC	Provides users with the ability to quickly query and analyze birth and death data sets from the National Vital Statistics System	De-identified demographic, geographic location, place, and cause information for deaths	Members of the public who are willing to agree to not use data to identify any individuals
WISQARS	CDC	Provides fatal and nonfatal injury, violent death, and cost of injury data from a variety of trusted sources, including the NVSS.	De-identified demographic, geographic, and cause information for injury-related deaths	Members of the public who are willing to agree to not use data to identify any individuals
Full DMF	SSA	Provides authorized user with access to all records of deaths reported to the SSA from 1936 to the present, including records reported to SSA by state and jurisdiction VROs through EDRS.	Name, social security account number, date of birth, and date of death of deceased individuals maintained by the Commissioner of Social Security	Federal benefits agencies, such as CMS, VA, OPM, and Federal Retirement Thrift Investment Board. See Table 7.
LADMF	SSA through NTIS	Provides certified users access to records of	Name, social security account	Individuals and organization who have successfully

System*	Responsible agency or organization	Description	Type of vital records data to which access is provided	Authorized users
		deaths reported to the SSA from 1936 to the present for (1) preventing identity fraud, (2) verifying death, and (3) identifying decedents in order to provide benefits to their beneficiaries. LADMF does not include protected state and jurisdiction vital records data.	number, date of birth, and date of death of deceased individuals maintained by the Commissioner of Social Security	completed the NTIS certification process. See Appendix D for a list of certified users.
EVVE	NAPHSIS	Provides users with the ability to quickly, reliably, and securely verify birth or death data with or without a paper certificate	None is provided to user. The system simply confirms or denies that there is a match of the individual in the request with a birth or death record.	State and federal government agencies
EVVE FOD	NAPHSIS	Provides users with the ability to quickly, reliably, and securely discover if a death record exists	Fact, place, and date of death, with other optional fields returned depending upon the jurisdiction	(1) federal benefits agencies, (2) federal administrative agencies, (3) state or local government benefits agencies, (4) state or local government administrative agencies, (5) non-governmental employers providing pension or retirement plans for their employees, (6) insurance companies, (7) companies “in the process of managing [their] assets,” and (8) institutions that deal with financial transactions, such as banks
NDI	CDC, NCHS	Assists investigators in determining whether persons in their research studies have died, and, if so, provides the names of the states in which those deaths occurred, the dates of death, and the corresponding death certificate numbers	Fact of death, jurisdiction, and death certificate number for identified deaths and, optionally, cause of death	Epidemiologists and other health and medical investigators for use in medical and health research

* State birth and death files usually can't be released under state statutes.

† Restricted data sets for research, including those with geographically identified data, are available through the NCHS Research Data Center (RDC). See <https://www.cdc.gov/rdc/index.htm> (access on 23 Oct 2017) for more information on the NCHS RDC.

†† Birth data includes demographic and medical information about the newborn and the mother.

Sources of information for Table 5:

- (1) CDC: CDC WONDER, available <https://wonder.cdc.gov>, accessed on 23 Oct 2017.
- (2) CDC: NCHS: Data User Agreement, available at https://www.cdc.gov/nchs/data_access/restrictions.htm, accessed on 23 Oct 2017.
- (3) CDC: Welcome to WISQARS, available at <https://www.cdc.gov/injury/wisqars/index.html>, accessed on 23 Oct 2017.
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- (11) Social Security: Requesting The Full Death Master File (DMF), available at https://www.ssa.gov/dataexchange/request_dmf.html; accessed on 17 October 2017.

Sources of revenue for vital records offices

Vital records offices potentially receive revenue from several sources, including state and local government appropriations; sale of birth and death certificates to individuals, families, and organizations; the NCHS' Vital Statistics Cooperative Program (VSCP); National Death Index use fees; SSA payments for birth and death data and records; payments from the NAPHSIS EVVE and EVVE FOD services; and payments from the Consumer Product Safety Commission for death certificates with specific causes of death (Table 6).

The ability of VROs to receive and use revenue from these sources varies by jurisdiction and by source. In some jurisdictions, virtually all revenue from the sale of birth and death certificates and vital records data goes into the jurisdiction's general fund, or to the agency that houses the VRO, such as the health department. The VROs in other jurisdictions are able to keep revenue from all sources to fund their operation, or to keep revenue from at least some sources.

Based on interviews for this report, there appears to be no complete listing by jurisdiction of the way in which vital records offices are funded, or how revenue from the sale of certificates and from NAPHSIS, NCHS, and SSA use of vital records data is handled administratively. There also appears to be no centralized compilation of the budgets and other financial information for the 57 U.S. vital registration jurisdictions, using standardized definitions and accounting methods.

Table 6. Sources of revenue for state and jurisdiction vital records offices (VRO)

1. Appropriations from state, local, or territorial government for VRO
2. Revenue from sale of birth and death certificates to individuals and families
 - a. # certificates issued x cost per certificate (see Appendix F for cost and availability of certificates by state or jurisdiction)
3. Revenue from sale of individual birth and death certificates to private organizations
 - a. # certificates issued x cost per certificate
4. Revenue from preparation of new birth certificates by the VRO for adoptions and changes in paternity information
 - a. # certificates prepared x cost per certificate
 - b. # certificates changed x cost per certificate
5. Revenue from amendments (e.g., name change) to birth certificates by the VRO
 - a. # certificates amended x cost per certificate
6. Revenue from state, local, or territorial government agency(ies) for providing access to birth or death records to carry out agency's mission (e.g., DMV, Medicaid, Injury Prevention)
7. Contracts with VROs under the NCHS Vital Statistics Cooperative Program (VSCP)
 - a. NAPHIS and NCHS negotiate the content and revenue for records that will be exchanged, and then NCHS pays states directly via contracts for their records.
 - b. **Total amount of VSCP for most recent year: \$21,148,548.** (Note: The total amount of the VSCP contract with state VROs has been constant for the past 5 years.)
 - c. The annual VSCP allocation to each jurisdiction is specified in the VSCP contract. Each jurisdiction receives at least \$50,000. All VSCP funds go to states or jurisdictions, but not all VROs are able to use these funds. In some jurisdictions, the funds go the jurisdiction's general fund, rather than to the jurisdiction's VRO.
8. Special projects funding of VROs through the NCHS VSCP
 - a. Project specific funding is awarded to states and territories on a competitive basis.
 - b. **Total amount of VSCP Special Project Funding awarded to jurisdictions for the 5-year period, 2013–2017 was \$10,962,969.** All 57 vital registration jurisdictions received at least some funding during this period, with median funding for jurisdictions of \$178,095 and a range of \$6,000 to \$689,037. See Appendix G.
9. Revenue from National Death Index
 - a. NDI total revenue in dollars and amount paid to jurisdictions for past 3 years:

Federal fiscal year	Total NDI revenue (\$)	NDI revenue paid to jurisdictions (\$)
2015	4,137,314	1,687,286
2016	4,305,463	1,685,269
2017	6,170,434	2,763,711

- b. NDI revenue varies by year depending on the number and size of the requests from researchers
 - c. Allocation formula for each state or jurisdiction is based on the number of death records matched by each state or jurisdiction: see description at https://www.cdc.gov/nchs/data/ndi/ndi_user_fees_worksheet.pdf.
10. Contract between VROs and SSA for electronic birth and death records to support SSA mission
 - a. SSA pays state, local, and territorial governments for birth and death records.
 - b. Allocation formula for each state and jurisdiction: # records provided x payment per record

- c. Current SSA payment: \$3.67 per birth record, and the following payments for death records, based on when SSA receives each record:

Timeframe for SSA receipt of death record	Payment per record
SSA receives EDR* within 6 business days of death	\$3.16
SSA receives EDR within 7–30 business days of death	\$1.83
SSA receives EDR or other death record within 31–120 business days of death	\$0.91
SSA receives EDR or other death record beyond 120 business days of death	\$0.01

* EDR = Electronic Death Record

- d. State registrars are proposing an increase in the SSA payment from \$3.16 to \$4.88 per EDR received within 6 business days of death.
- e. Based on current payment formulas listed above, birth and death data for 2015 from NCHS, ^{7,8} and the assumption that SSA received an EDR within 6 business days for all 2015 deaths, the estimated total SSA payments to VROs for 2015 would be the following:

(1) Birth records: \$3.67 per record x 3,978,497 births $\cong$ \$14,601,084

(2) Death records: \$3.16 per record x 2,712,630 deaths $\cong$ \$8,571,911

11. Revenue from NAPHSIS EVVE service

- a. **EVVE queries currently produce revenues of about \$1.2 million per year.**
- b. NAPHSIS reimburses each state or jurisdiction from revenues it receives from EVVE users.
- c. Allocation formula for each state or jurisdiction: (# births and deaths matched x cost per match) + (# certifications issued x cost per certification)

12. Revenue from NAPHSIS EVVE FOD service

- a. NAPHSIS reimburses each state or jurisdiction from revenues it receives from EVVE FOD users after NAPHSIS expenses for developing and managing EVVE FOD have been deducted.
- b. Allocation formula for each state or jurisdiction: (EVVE FOD revenue - NAPHSIS expenses) x % total deaths occurring in the jurisdiction x participation amount for all jurisdictions.

13. Revenue from Consumer Product Safety Commission's purchase of death certificates with cause of death codes for which there is a high probability that consumer products are involved.

14. Revenue from the private sector (e.g., Ancestry.com) for use of records as a resource (only available in a few states)

Notes:

- (1) Depending on the state or jurisdiction, revenue from the sale of certificates or other VRO registration services may go to support the VRO directly, or it may go into the general fund of the health department or the state government. In some states, revenue from the sale of

⁷ Martin JA, Hamilton BE, Osterman MJK, et al. Births: Final data for 2015. National vital statistics report; vol 66, no 1. Hyattsville, MD: National Center for Health Statistics. 2017.

⁸ Xu JQ, Murphy SL, Kochanek KD, Arias E. Mortality in the United States, 2015. NCHS data brief, no 267. Hyattsville, MD: National Center for Health Statistics. 2016.

certificates may go to local issuing offices or agents (e.g., Vermont).

- (2) Revenue from sales may be carried over to future years in some states (e.g., NH), but in other states the revenues must be spent by the VRO before the end of the fiscal year, or be lost.

Sources of information for Table 6:

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- (2) NAPHSIS: About EVVE. Available at <https://www.naphsis.org/evve>; accessed on 4 Oct 2017.
- (3) Personal communication with Delton Atkinson, Director, Division of Vital Statistics, NCHS, CDC, 24 Oct 2017.
- (4) Personal communication with Rich McCoy, Vermont State Registrar and Chief, Center for Health Statistics, 13 Oct 2017.
- (5) Personal communication with Shawna Webster, Executive Director, NAPHSIS, 5 Oct 2017.
- (6) Presentations at NCVHS hearing, September 11-12, 2017, Washington, D.C., available at <https://www.ncvhs.hhs.gov/meeting-calendar/agenda-of-the-september-11-12-2017-subcommittee-on-population-health-next-generation-vital-statistics-hearing/>; accessed on 4 Oct 2017.
- (7) Rothwell CJ (CDC/NCHS), *SSA Death Master File & NCHS National Death Index: How do they relate?* [Slide Presentation], available at <https://www.cdc.gov/nchs/data/bsc/rothwell.pdf>; accessed on 9 Oct 2017.
- (8) U.S. Consumer Product Safety Commission: *Guide to Public Information: Death Certificate File*, available at <https://www.cpsc.gov/Newsroom/FOIA/Guide-to-Public-Information>; accessed 6 Nov 2017.

Conclusions

This report summarizes the uses, users, systems, and sources of revenue for vital records and vital statistics in the United States. Vital records and vital statistics are critical for many legal, administrative, financial, policy, health research, and public and private benefit program needs. In spite of their importance, relatively little is known about the actual functioning, funding, and long-term stability and viability of the vital records offices and vital statistics programs—both individually and as a whole—that collect, manage, archive, and make available vital records and statistics and constitute the national vital records and statistics system. Many public and private users appear to benefit from vital records and statistics, but relatively few of them appear to directly fund the vital records and statistics system. To better understand this system and to aid in undertaking future improvements of it, the author of this report recommends a systematic effort to obtain the following information for each state and jurisdiction:

- (1) Costs for running VROs. Information should be systematically collected using a standardized approach, including standardized definitions and expense categories.
- (2) Revenues from sale of birth and death certificates for each state/jurisdiction
- (3) Agencies/programs that receive and have authority to spend revenues generated by vital certificate sales
- (4) Distribution of (a) funds from VSCP contracts, (b) sale of birth and death data to SSA, and (c) revenue from other federal agencies
- (5) Distribution of revenues from the use of birth and death data by state agencies
- (6) Laws and statutes that control/limit the ability of VROs to distribute/sell identified birth and death data to federal and state government agencies and to private parties and organizations
- (7) Satisfaction of users of vital records and vital statistics with VRO services, including the timeliness and cost of obtaining vital records; and user suggestions for improving the current system to better meet their needs

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- (4) CDC: National Center for Health Statistics. Accessing Data from the National Center for Health Statistics, available at https://www.cdc.gov/nchs/data_access/index.htm; accessed 6 Nov 2017.
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- (6) CDC: National Center for Health Statistics. Resources for Researchers, available at https://www.cdc.gov/nchs/nchs_for_you/researchers.htm; accessed 6 Nov 2017.
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Appendices

Appendix A. Abbreviations and acronyms used in report

CDC: Centers for Disease Control and Prevention
CMS: Centers for Medicare and Medicaid Services
CPSC: Consumer Products Safety Commission
DHS: United States Department of Homeland Security
DMF: Death Master File
DMV: Department of Motor Vehicles
DoD: United States Department of Defense
DoL: United States Department of Labor
DoS: United States Department of State
EBRS: Electronic Birth Registration System
EDRS: Electronic Death Registration System
EVVE: Electronic Verification of Vital Events
EVVE FOD: Electronic Verification of Vital Events Fact of Death
FBI: Federal Bureau of Investigation
FSA: United States Department of Agriculture's Farm Service Agency
IIS: Immunization Information Systems
IRS: Internal Revenue Service
LADMF: Limited Access Death Master File
NAPHSIS: National Association of Public Health Statistics and Information Systems
NCHS: National Center for Health Statistics
NDI: National Death Index
NRCS: United States Department of Agriculture's Natural Resources Conservation Service
OPM: United States Office of Personnel Management
OWCP: United States Department of Labor's Office of Workers' Compensation Programs
RMA: United States Department of Agriculture's Risk Management Agency
SSA: Social Security Administration
STEVE: State and Territorial Exchange of Vital Events system
USDA: United States Department of Agriculture
USCIS: United States Department of Homeland Security's United States Citizenship and Immigration Services
VA: U.S. Department of Veterans Affairs
VRO: Vital Records Office
VSCP: Vital Statistics Cooperative Program

Appendix B. Brief descriptions of selected uses and users listed in tables and figures

Citizenship and Immigration Services. The U.S. Citizenship and Immigration Services (USCIS), which is located within the U.S. Department of Homeland Security, is the government agency that oversees lawful immigration to the United States. The USCIS has 19,000 government employees and contractors working at 223 offices across the world. USCIS administers the program that verifies that an individual may legally work in the United States by comparing information from an employee's Form I-9 with other government records. USCIS is funded primarily by immigration and naturalization benefit fees charged to applicants and petitioners. Source: U.S. Department of Homeland Security: Citizenship and Immigration Services, available at <https://www.state.gov/m/ds/rls/274294.htm>, accessed on 23 Oct 2017.

Defense Manpower Data Center. Since 1974, the DMDC has evolved into a world leader in Department of Defense (DoD) identity management, serving uniformed service members and their families across the globe. Among its many roles, DMDC is:

- The leader in joint information sharing and support on DoD human resource issues
- The central source for identifying, authenticating, authorizing, and providing information on personnel during and after their affiliation with DoD
- The one, central access point for information and assistance on DoD entitlements, benefits, and medical readiness for uniformed service members, veterans, and their families.

Source: U.S. Department of Defense: Defense Manpower Data Center, available at <https://www.dmdc.osd.mil/appj/dwp/index.jsp>, accessed on 23 Oct 2017.

Diplomatic Security. The Diplomatic Security Service (DSS) is the law enforcement and security arm of the U.S. Department of State. It bears the core responsibility for providing a safe environment for the conduct of U.S. foreign policy. DSS may use birth certificates and birth data to identify adoption fraud, which may rely on document fraud, including fraudulent U.S. passports and visas. Sources: (1) U.S. Department of State: Bureau of Diplomatic Security, available at <https://www.state.gov/m/ds/>, accessed on 23 Oct 2017; (2) U.S. Department of State: Diplomacy and Law Enforcement Unite to Dismantle Major International Adoption Fraud Scheme, by Angela French, DSS Public Affairs, available at <https://www.state.gov/m/ds/rls/274294.htm>, accessed on 23 Oct 2017.

Federal Workers' Compensation. The Department of Labor's Office of Workers' Compensation Programs (OWCP) administers four major disability compensation programs, which provide to federal workers (or their dependents) who are injured at work or acquire an occupational disease the following benefits: (1) Wage replacement benefits, (2) Medical treatment, (3) Vocational rehabilitation, and (4) Other benefits. The following compensation programs cover other specific groups: (1) Energy Employees Occupational Illness Compensation Program, (2) Federal Employees' Compensation Program, (3) Longshore and Harbor Workers' Compensation Program, and (4) Black Lung Benefits Program. These entities serve the specific employee groups who are covered under the relevant statutes and regulations by mitigating the financial burden resulting from workplace injury. Source: United States Department of Labor: Workers' Compensation, available at <https://www.dol.gov/general/topic/workcomp>; accessed on 4 Oct 2017.

Federal Retirement Thrift Investment Board. The Federal Retirement Thrift Investment Board administers the Thrift Savings Plan (TSP), a tax-deferred defined contribution plan similar to private sector 401(k) plans, which provides Federal employees and members of the uniformed services the

opportunity to save for additional retirement security. Source: Federal Retirement Thrift Investment Board: Home, available at <https://www.frtib.gov>; accessed on 9 Oct 2017.

Immunization Information Systems (IIS). Previously known as immunization registries, these are state-based programs that maintain “confidential, population-based, computerized databases that record all immunization doses administered by participating providers to persons residing within a given geopolitical area.” The IIS in most states works with the state VRO to establish new immunization records for children born and residing in the state. See NCVHS hearing testimony from the American Immunization Registry Association (AIRA). See also AIRA at <http://www.immregistries.org>, accessed on 30 Sep 2017; and CDC: Immunization Information Systems, available at <https://www.cdc.gov/vaccines/programs/iis/index.html>, accessed on 30 Sep 2017.

Newborn screening. State-based programs that screen newborns for a number of genetic, endocrine, and metabolic disorders; critical congenital heart defects; and hearing loss. Source: (1) CDC: Newborn Screening Portal, available at <https://www.cdc.gov/newbornscreening/>, accessed on 30 Sep 2017; and (2) individual state newborn screening program websites, available at <http://www.savebabies.org/screening.html>, accessed on 30 Sep 2017.

Office of Workers' Compensation Programs. The U.S. Department of Labor’s Office of Workers' Compensation Programs administers four major disability compensation programs which provide wage replacement benefits, medical treatment, vocational rehabilitation and other benefits to certain workers or their dependents who experience work-related injury or occupational disease:

- Energy Employees Occupational Illness Compensation program,
- Federal Employees' Compensation Program,
- Longshore and Harbor Workers' Compensation Program, and
- Coal Mine Workers' Compensation Program.

Source: United States Department of Labor: Office of Workers' Compensation Programs (OWCP), available at <https://www.dol.gov/owcp/owcpabot.htm>, accessed on 23 Oct 2017.

Passport Services. The Department of State's Passport Services Directorate issues U.S. passports to traveling Americans. It protects the integrity of the U.S. passport as proof of U.S. citizenship at home and around the world. Applicants for a U.S. Passport must provide proof of U.S. citizenship, consisting of a previous U.S. passport, certified birth certificate, or other documentation acceptable to Passport Services. Source: U.S. Department of State, Bureau of Consular Affairs: U.S. Passports & International Travel, available at <https://travel.state.gov/content/passports/en/passports.html>, accessed on 23 Oct 2017.

Pension Benefit Guaranty Corporation. PBGC guarantees the "basic benefits" that people earn before their pension plan’s termination date (or the date their employer’s bankruptcy proceeding began, if applicable) up to legal limits set by Congress. PBGC pays benefits:

- If a sponsoring company seeking to reorganize in bankruptcy proves that it cannot remain in business and continue funding the pension plan
- If a plan runs out of money to pay benefits due
- If a sponsoring company files for liquidation (as opposed to reorganization)

Source and additional information is available at Pension Benefit Guaranty Corporation: A U.S. Government Agency, <https://www.pbgc.gov>; accessed on 9 Oct 2017.

U.S. Railroad Retirement Board. The Railroad Retirement Board (RRB) is an independent agency in the

executive branch of the Federal Government. The RRB's primary function is to administer comprehensive retirement-survivor and unemployment-sickness benefit programs for the nation's railroad workers and their families, under the Railroad Retirement and Railroad Unemployment Insurance Acts. As part of the retirement program, the RRB also has administrative responsibilities under the Social Security Act for certain benefit payments and railroad workers' Medicare coverage. Source and additional information is available at U.S. Railroad Retirement Board: Menu: Agency Overview, <https://www.rrb.gov>; accessed on 9 Oct 2017.

Appendix C. Descriptions of selected information systems based on vital records that provide data about vital events to various users

Death Master File (DMF)

The Social Security Administration (SSA) collects death information to administer its programs. Death reports come from many sources, including family members, funeral homes, financial institutions, postal authorities, states and other federal agencies, but the death reports collected by SSA are NOT a comprehensive record of all deaths in the United States.

The SSA compiles its death information in the Death Master File (DMF), which is an extract of death information on NUMIDENT, the electronic database that contains SSA's records of Social Security Numbers (SSN) assigned to individuals since 1936. The DMF includes, if available, each deceased individual's SSN, first name, middle name, surname, date of birth, and date of death. SSA currently prepares two versions of the DMF:

- (1) The full DMF contains all death records extracted from the SSA NUMIDENT database, including death data received from states, and is shared only with certain federal and state agencies pursuant to section 205(r) of the Social Security Act.⁹

⁹ Section (r)(1): "The Commissioner of Social Security shall undertake to establish a program under which—

(A) States (or political subdivisions thereof) voluntarily contract with the Commissioner of Social Security to furnish the Commissioner of Social Security periodically with information (in a form established by the Commissioner of Social Security in consultation with the States) concerning individuals with respect to whom death certificates (or equivalent documents maintained by the States or subdivisions) have been officially filed with them; and

(B) there will be (i) a comparison of such information on such individuals with information on such individuals in the records being used in the administration of this Act, (ii) validation of the results of such comparisons, and (iii) corrections in such records to accurately reflect the status of such individuals."

Section (r)(2): "Each State (or political subdivision thereof) which furnishes the Commissioner of Social Security with information on records of deaths in the State or subdivision under this subsection may be paid by the Commissioner of Social Security from amounts available for administration of this Act the reasonable costs (established by the Commissioner of Social Security in consultations with the States) for transcribing and transmitting such information to the Commissioner of Social Security."

Section 205(r)(3): "In the case of individuals with respect to whom federally funded benefits are provided by (or through) a Federal or State agency other than under this Act, the Commissioner of Social Security shall to the extent feasible provide such information through a cooperative arrangement with such agency, for ensuring proper payment of those benefits with respect to such individuals if—(A) under such arrangement the agency provides reimbursement to the Commissioner of Social Security for the reasonable cost of carrying out such arrangement, and (B) such arrangement does not conflict with the duties of the Commissioner of Social Security under paragraph (1)." [Emphasis added.]

Section 205(r)(6): Information furnished to the Commissioner of Social Security under this subsection may not be used for any purpose other than the purpose described in this subsection and is exempt from disclosure under section 552 of title 5, United States Code, and from the requirements of section 552a of such title.

Section 205(r)(8)(A): The Commissioner of Social Security shall, upon the request of the official responsible for a State driver's license agency pursuant to the Help America Vote Act of 2002— (i) enter into an agreement with such official for the purpose of verifying applicable information, so long as the requirements of subparagraphs (A) and (B) of paragraph (3) are met; and ... (ii) include in such agreement safeguards to assure the maintenance of the confidentiality of any applicable information disclosed and procedures to permit such agency to use the applicable information for the purpose of maintaining its records.

Source: Social Security: Compilation of the Social Security Laws: Evidence, Procedure, and Certification for Payment: Sec. 205. [42 U.S.C. 405], available at https://www.ssa.gov/OP_Home/ssact/title02/0205.htm; accessed

- (2) The public DMF contains death records extracted from the SSA NUMIDENT database, but does not include death data received from the states. SSA provides this version to the Department of Commerce's National Technical Information Service, a clearinghouse for government information, which sells it to the public (other agencies and private organizations such as banks and credit companies) as the Limited Access Death Master File (LADMF).¹⁰

The LADMF has three principal uses: (1) prevent identity fraud, (2) verify death, and (3) identify decedents in order to provide benefits to their beneficiaries, which are described in more detail below.

(1) By methodically running financial, credit, payment and other applications against the LADMF, the financial community, insurance companies, security firms and state and local governments are better able to identify and prevent identity fraud.

(2) The LADMF is important for death verification. Medical researchers, hospitals, oncology programs all need to track former patients and study subjects. Investigative firms use the data to verify the death of persons, in the course of their investigations.

(3) "Pension funds, insurance organizations, Federal, State and Local governments and others responsible for payments to recipients/retirees all need to know if they might be sending checks to deceased persons. Individuals may search for loved ones, or work toward growing their family trees. Professional and amateur genealogists can search for missing links." See Social Security Death Master File (www.ssdmf.com/; accessed on 4 Oct 2017). "The Death Master File is an important tool which can be used by pension funds, insurance organizations, Federal, State and Local governments and others responsible for verifying deceased person(s) in support of fulfillment of benefits to their beneficiaries."

As described above, the LADMF is available from the National Technical Information Service (NTIS). A user may access an online search application, or download and maintain a raw data version of the file. The online service is updated weekly, and the weekly and monthly updates are offered electronically via https to reduce handling and production time. NTIS established a certification program for those seeking access to the LADMF pursuant to Section 203 of the Bipartisan Budget Act of 2013 (Pub. L.113-67);¹¹ the

on 9 Oct 2017.

¹⁰ The public DMF, which was created in 1980 in response to a 1978 FOIA lawsuit, originally contained over 89 million records of deaths reported to SSA since 1936. In 2002, the SSA public DMF began disclosing state electronic death records. In 2010–2011, SSA examined its disclosure of "protected" state death records, and determined that Section 205(r) of the Social Security Act prohibits SSA from disclosing the state death records that SSA receives through its contracts with the states, except in limited circumstances. As of November 1, 2011, SSA decided to withdraw protected state death records from the public DMF data made available to NTIS for online search and download products. The public DMF continued to contain non-state records, but the number of death records in the public DMF was reduced by approximately 4.2 million, and SSA now adds about 1 million fewer records annually to the DMF. The public DMF, which is distributed by NTIS as the LADMF, currently contains over 86 million records. A user may access an online search application, or download and maintain a raw data version of the file. The online service is updated weekly, and the weekly and monthly updates are offered electronically via https to reduce handling and production time.

¹¹ Section 203 of the Bipartisan Budget Act of 2013, Public Law 113-67 (Act), passed into law on December 26, 2013 prohibits the Secretary of Commerce from disclosing DMF information during the three-calendar- year period following an individual's death (referred to as the "Limited Access DMF," or "LADMF"), unless the person requesting the information has been certified to access that information pursuant to certain criteria in a program that the Secretary establishes. The Act further requires the Secretary to establish a fee-based program to certify Persons for access to LADMF. In addition, it provides for penalties for Persons who receive or distribute LADMF without being certified or otherwise satisfying the requirements of the Act. The Secretary delegated the authority to carry out Section 203 to the Director of NTIS. The final rule for the certification program became effective

program limits access to LADMf information to those persons certified under the program.¹² Certified persons, also called Subscribers, must have a legitimate fraud prevention interest, or have a legitimate business purpose pursuant to a law, governmental rule, regulation, or fiduciary duty in order to be certified under the program. Section 203 requires that NTIS charge fees sufficient to cover the costs associated with the certification program. Information on the cost of using the DMF is available from NTIS at NTIS: Limited Access Death Master File Available Through Value-Added Online Products (<https://classic.ntis.gov/products/ssa-online/>). A list of current certified users can be found in Appendix D and here: <https://classic.ntis.gov/assets/pdf/DMFcertifiedList.docx>.

Table 7. Death Master File death records exchanges conducted under Section 205(r) of the Social Security Act, as of 25 August 2017

Benefit-Paying Agencies	Version	Frequency
Centers for Medicare and Medicaid Services	Public+State	Weekly updates
Dept. of Agriculture—Farm Service Agency, Natural Resources Conservation Service, Risk Management Agency	Public+State	Weekly updates & 1 full file in FY 16
Dept. of Defense—Defense Manpower Data Center	Public+State	Monthly updates
Dept. of Housing and Urban Development	Public+State	Weekly & 1 full file in June 2017
Dept. of Veterans Affairs	Public+State	Weekly updates & 1 full file annually
Federal Retirement Thrift Investment Board	Public+State	Monthly updates
Internal Revenue Service	Public+State	Weekly updates & 1 full file annually
Office of Personnel Management	Public+State	Weekly updates & 1 full file annually
Pension Benefit Guaranty Corporation	Public+State	Weekly updates & 1 full file annually
Railroad Retirement Board	Public+State	Monthly updates
Non-Benefit-Paying Agencies		
Government Accountability Office	Public+State	Monthly updates
National Technical Information Service	Public	Monthly & weekly updates & 1 full file quarterly

November 28, 2016.

¹² Executive departments or agencies of the United States Government are not considered “Persons” for the purposes of this rule; and Executive departments or agencies do not have to complete the Certification Form as set forth in the rule, and can access Limited Access DMF under a subscription or license agreement with NTIS, describing the purpose(s) for which Limited Access DMF is collected, used, maintained and shared. Those working on behalf of and authorized by Executive departments or agencies may access the Limited Access DMF from their sponsoring Executive department or agency, which will be responsible for ensuring that such access is solely for the authorized purposes described by the agency. Unauthorized secondary use of Limited Access DMF by Executive departments or agencies or those working for them or on their behalf is prohibited. If an Executive department or agency wishes those working on its behalf to access the Limited Access DMF directly from NTIS, then those working on behalf of that Executive department or agency are required to complete and submit the Certification Form as set forth in the rule and enter into a subscription agreement with NTIS in order to directly access the Limited Access DMF.

Source for information in Table 7: Email with Subject: Users of the 205(r) data: DMF Customers and Death Exchanges, from Tom Klouda, U.S. Senate, August 25, 2017 at 2:19:10 PM EDT.

Sources of information about the Death Master File:

- (1) Department of Commerce, National Technical Information Service, Certification Program for Access to the Death Master File. Federal Register, June 1, 2016, Volume 81, Number 105, pages 34882-34895; available at <https://www.gpo.gov/fdsys/pkg/FR-2016-06-01/html/2016-12479.htm>.)*
- (2) Department of Commerce, National Technical Information Service. List of persons certified to receive the Limited Access Death Master File as of June 1, 2017, available at <https://classic.ntis.gov/assets/pdf/DMFcertifiedList.docx>; accessed on 4 Oct 2017.*
- (3) NAPHSIS: About EVVE FOD, available at <https://www.naphsis.org/evve>; accessed on 4 Oct 2017.*
- (4) NTIS: Limited Access Death Master File (<https://classic.ntis.gov/products/ssa-dmf/#>); accessed on 4 Oct 2017).*
- (5) NTIS: Limited Access Death Master File Available Through Value-Added Online Products, available at <https://classic.ntis.gov/products/ssa-online/>; accessed on 4 Oct 2017.*
- (6) Rothwell CJ (CDC/NCHS), SSA Death Master File & NCHS National Death Index: How do they relate? [Slide Presentation], available at <https://www.cdc.gov/nchs/data/bsc/rothwell.pdf>; accessed on 9 Oct 2017.*
- (7) Social Security Death Master File, available at www.ssdmf.com/; accessed on 4 Oct 2017.*
- (8) Social Security: Compilation of the Social Security Laws: Evidence, Procedure, and Certification for Payment: Sec. 205. [42 U.S.C. 405], available at https://www.ssa.gov/OP_Home/ssact/title02/0205.htm; accessed on 9 Oct 2017.*
- (9) Social Security: Requesting The Full Death Master File (DMF), available at https://www.ssa.gov/dataexchange/request_dmf.html; accessed on 17 October 2017.*
- (10) U.S. Department of Commerce: National Technical Information Service. Important Notice: Change in Public Death Master File Records. Available at <https://classic.ntis.gov/assets/pdf/import-change-dmf.pdf>; accessed on 4 Oct 2017.*

Electronic Verification of Vital Events (EVVE)

The Electronic Verification of Vital Events, which became operational in 2002 and is operated by NAPHSIS, provides customers with the ability to quickly, reliably, and securely verify birth or death data with or without a paper certificate. State agencies (e.g., DMV, Secretary of State, Medicaid, and Army National Guard) and federal agencies (e.g., SSA, OPM, and Dept of State) actively utilize EVVE to improve business operations and customer service, and to protect against fraudulent activities. Currently, EVVE has about 20 customers, including the following:

- Social Security Administration
- Office of Personnel Management
- Department of Motor Vehicles
- Medicaid Offices
- Department of Homeland Security - USCIS
- Department of State - Passport Services Fraud Prevention Offices
- Department of State - Diplomatic Security
- Army National Guard
- Regional FBI Offices
- Secretary of State Offices

EVVE currently has access to over 250 million birth and death records from state and jurisdiction owned

vital record databases nationwide. (Currently, all states except New York and Texas fully participate in EVVE birth and death verification. All territories participate in EVVE birth verification, but only Puerto Rico participates in EVVE death verification, as the other territories do not currently have electronic death registration.)

Electronic inquiries from authorized users—inquiries that are usually derived from information about a person from a vital record, form, or other document that the user wishes to verify—are sent to the participating state(s) and jurisdiction(s) specified in the query.¹³ The EVVE response to a query notifies the user that the person of interest either matches or does not match an EVVE vital record. If the query matches a record, the user has the choice of obtaining official certification of the record from the relevant state or jurisdiction's vital records office. See Figure 3 below. Most EVVE queries (>97%) are for birth verifications; death verifications, which make up less than 3% of EVVE queries, are more commonly done through EVVE FOD, which is described below.

Figure 3. Steps in the Electronic Verification of Vital Events system for verifying birth and death information



Source: NAPHSIS: About EVVE. Available at <https://www.naphsis.org/evve>; accessed on 4 Oct 2017.

EVVE users are charged for queries based on the type of query. A verification query is done when the user has a certificate in-hand, and the query requires data from that certificate. A \$0.55 basic query fee is charged; and if a match is obtained, a \$1.35 jurisdiction fee is charged as well. A certification query is done when the user does not have a certificate; this type of query provides an Electronic Birth Certificate or an Electronic Death Certificate. As such, the jurisdiction fee is the same as the cost for a paper certificate from that jurisdiction; and there is an additional EVVE system fee, which is 18% of the jurisdiction fee in addition to the \$0.55 basic query fee.

EVVE queries currently produce revenues of about \$1.2 million per year. Approximately 80% of the basic query fee goes to the state with the vital record for the person of interest to the user; the remainder of the basic query fee goes to NAPHSIS to support the operation of EVVE.

Source of information about EVVE:

- (1) Email communication with NAPHSIS staff Anthony Stout on 16 Oct 2017.
- (2) NAPHSIS: About EVVE. Available at <https://www.naphsis.org/evve>; accessed on 4 Oct 2017;

¹³ As part of a future EVVE update, if the user does not know the jurisdiction for the vital event, EVVE will have the ability to “broadcast” the query to all participating EVVE states and jurisdictions.

Electronic Verification of Vital Events Fact of Death (EVVE FOD)

EVVE FOD, which became operational in March 2017 and is owned and operated by NAPHSIS and managed by LexisNexis VitalChek, provides customers with the ability to quickly, reliably, and securely discover if a death record exists. EVVE FOD provides an alternative to the Security Administration's (SSA) Death Master File (DMF) for authorized users to obtain information on the fact of death. (To be authorized, a user must be from one of the 8 following organizational categories: (1) federal benefits agencies, (2) federal administrative agencies, (3) state or local government benefits agencies, (4) state or local government administrative agencies, (5) non-governmental employers providing pension or retirement plans for their employees, (6) insurance companies, (7) companies "in the process of managing [their] assets," and (8) institutions that deal with financial transactions, such as banks. Currently, EVVE FOD has about 11 users.) Currently, 42 vital registration jurisdictions are fully participating in the EVVE FOD system by providing access to authorized categories of users to current, accurate, and complete records of deaths that occur in their jurisdictions. (See Figure 4 for a map showing fully participating states and jurisdictions. *Note that access to death records in these participating states and jurisdictions may vary by the category of user. For example, access for federal agencies providing benefits may differ from the access provided to insurance companies.*)

The EVVE FOD user can submit a single query or a group of queries, usually through the EVVE FOD web page. User queries are sent to all participating states and jurisdictions—based on the user's business category—to be matched against death records from state and jurisdiction owned vital record databases. (Schedule A of the Regulatory Settlement Agreements between state insurance departments and insurance companies specifies the criteria used by EVVE FOD for matching. The criteria include exact Social Security Number (SSN) matches and "fuzzy" matches based on first and last name, date of birth, and SSN.) Once the query has been completed, any matches found will return the date of death as well as the place of death. Depending on where—i.e., the state or jurisdiction—the event occurred, more or less detailed information may be returned.

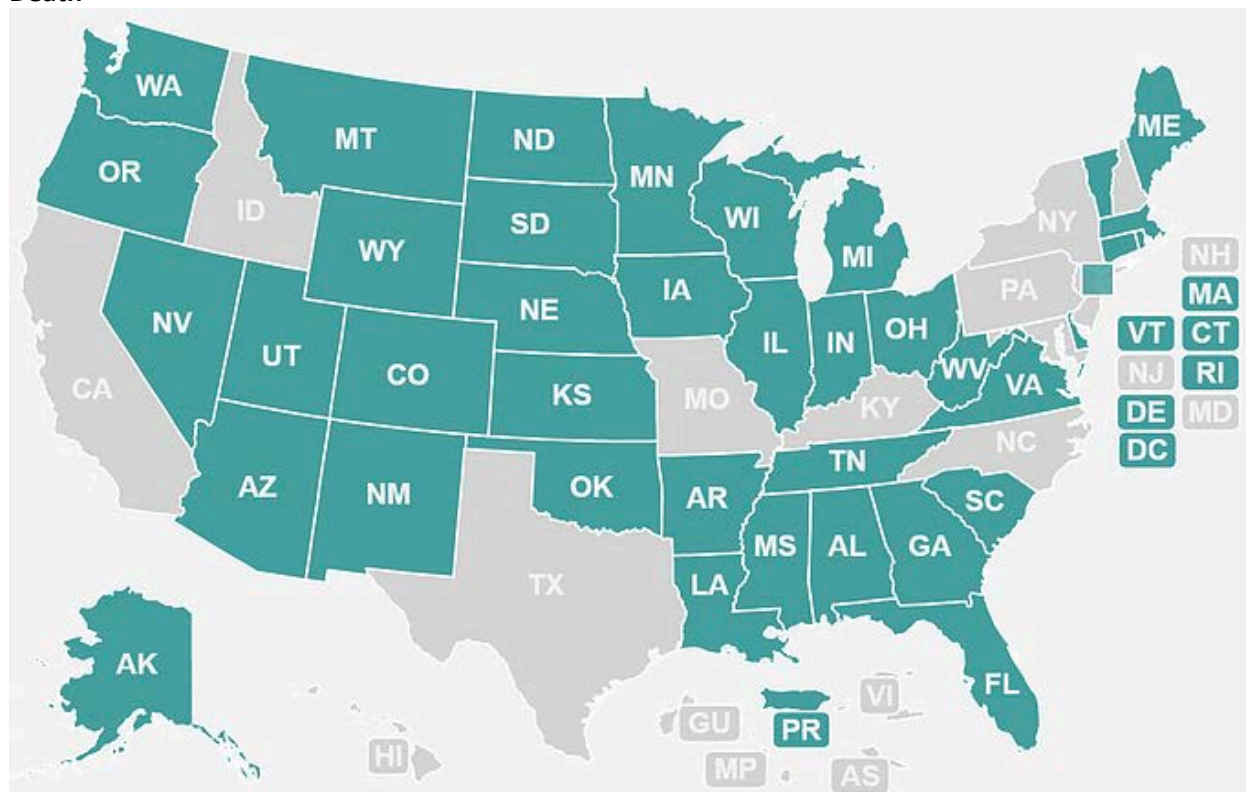
EVVE FOD users are charged based on (1) the type of user: private (full price), federal government (30% discount), or state or local government (60% discount); and (2) the number of queries submitted per month.¹⁴ NAPHSIS used funds from its "investment reserves" to develop the EVVE FOD product, and revenues from EVVE FOD queries will go to NAPHSIS until development costs have been covered. At that point, net revenue after expenses will be split 50-50 between NAPHSIS and the jurisdictions. How much a jurisdiction participates, its proportionate number of occurrence deaths, and the ability of other jurisdictions to participate will determine the jurisdiction's portion of the net revenues.

Sources of information about EVVE FOD:

- (1) *Email communication with NAPHSIS staff Anthony Stout on 16 Oct 2017, 15 Nov 2017, and 16 Nov 2017.*
- (2) *NAPHSIS: About EVVE FOD. Available at <https://www.naphsis.org/evve-fod>; accessed on 4 Oct 2017.*

¹⁴ The pricing structure for EVVE FOD is available at NAPHSIS: About EVVE FOD: EVVE FOD Questions? Available at <https://www.naphsis.org/evve-fod>; accessed on 6 Oct 2017.

Figure 4. States and jurisdictions fully participating in the Electronic Verification of Vital Events Fact of Death



Participating jurisdictions are shown in green and include New York City. Although not shown on the map, Maryland also participates in EVVE FOD, raising the number of participating jurisdictions to 42. Note that access to death records in these participating states and jurisdictions may vary by the type of user. For example, access for federal agencies providing benefits may differ from the access provided to insurance companies.

Sources:

(1) NAPHIS: *About EVVE FOD: FOD Use Maps*. Available at: <https://www.naphsis.org/evve-fod>, accessed on 6 Oct 2017

(2) email communication with NAPHIS staff Anthony Stout on 15 Nov 2017.

National Death Index (NDI)

“The NDI is a centralized database of death record information on file in state vital statistics offices. Working with these state offices, the National Center for Health Statistics (NCHS) established the NDI as a resource to aid epidemiologists and other health and medical investigators with their mortality ascertainment activities.

- **Assists investigators in determining whether persons in their studies have died** and, if so, provide the names of the states in which those deaths occurred, the dates of death, and the corresponding death certificate numbers. Investigators can then make arrangements with the appropriate state offices to obtain copies of death certificates or specific statistical information such as cause of death. **Cause of death codes may also be obtained using the NDI Plus service.**
- **Records from 1979 through 2015** are available and contain a standard set of identifying information on each death. Death records are added to the NDI file annually, approximately 12 months after the end of a particular calendar year.

- The NDI service is **available to investigators solely for statistical purposes in medical and health research**. The service is *not* accessible to organizations or the general public for legal, administrative, or genealogy purposes.”

The NDI Retrieval Program is used to search the NDI file to determine whether a particular NDI death record qualifies as a possible record match with a particular user record. To qualify as a possible record match, both records must satisfy at least one of seven conditions or matching criteria, and the specified data items must agree on both records. Chapter 4 of the NDI User Guide provides a complete list of the matching criteria. To facilitate matching, NDI users are encouraged to submit as many of the following data items as possible for each study subject: first and last name, middle initial, father's surname, social security number, month, day, and year of birth, race, sex, marital status, state of residence, and state of birth.

Researchers can request either a *routine* NDI search for those who do *not* need cause of death codes, or an NDI *Plus* search, which also provides cause of death codes. The cost for using the NDI consist of (1) a service charge (\$350 for initial submission and \$100 for each subsequent submission); (2) a cost per subject with unknown vital status x the number of years searched (\$0.21 per subject per year for NDI *Plus* or \$0.15 per subject per year for NDI *Routine*); and (3) a fixed cost for each subject known to be deceased (\$5.00 when researcher has no death certificate or \$2.50 when researcher has death certificate).¹⁵

Source of information above and additional information on the NDI are available at CDC: NCHS: National Death Index, <https://www.cdc.gov/nchs/ndi/index.htm>; accessed on 9 Oct 2017.

State and Territorial Exchange of Vital Events system (STEVE)

NAPHSIS began planning the State and Territorial Exchange of Vital Events system (STEVE) in spring 2006 with conversations about the potential for an information exchange system and how it could be funded. A meeting to gather functional requirements was held October 31–November 1, 2006. STEVE became operational in fall 2008 and allowed vital records jurisdictions to electronically complete the following transactions:

- (1) sending statistical data to the National Center for Health Statistics for inclusion in the National Vital Statistics System,
- (2) sending vital records that pertain to residents in other jurisdictions so the home state's reports include these important data,
- (3) sending death information to the jurisdiction of birth so that birth certificates can be flagged as 'deceased,' an important step in preventing fraud and identity theft, and
- (4) providing data to authorized data partners for use in authorized public health and administrative purposes, such as newborn hearing screening, cancer registries, the National Violent Death Registration System, and voter registration.¹⁶

The administrative foundation for the exchange among jurisdictions and authorized data partners is a legal agreement called the Interjurisdictional Exchange of Vital Records (IJE) Agreement, which establishes what data a state or jurisdiction will share and will not share. The IJE Agreement allows a

¹⁵ Costs were effective as of October 1, 2004. Additional information on NDI user fees is available at https://www.cdc.gov/nchs/data/ndi/ndi_user_fees_worksheet.pdf; accessed 9 Oct 2017.

¹⁶ The list of authorized data partners is included in the Interjurisdictional Exchange of Vital Records (IJE) Agreement.

jurisdiction to share any of the following six IJE layouts with any other resident jurisdiction and NCHS:

- (1) natality
- (2) mortality
- (3) mortality roster
- (4) fetal death
- (5) birth-infant death
- (6) induced termination of pregnancy

STEVE is the mechanism by which the IJE Agreement is operationalized. STEVE was a software application that was installed in every vital records jurisdiction, most often on a server within the domain of what became a centralized information technology (IT) department. The software encrypted and sent data to NCHS, other jurisdictions, and the authorized data partners mentioned above. Specifically, required data for IJE layouts were extracted from a jurisdiction's electronic birth, infant death, and death registration systems and then transferred into STEVE. STEVE then distributed the first section of each layout to NCHS¹⁷; next, it parsed the remaining data according to the rules of the IJE Agreement and distributed appropriate data to other jurisdictions and data partners. Upon arrival at the recipient's electronic mailbox, data files were decrypted by STEVE and could be opened in Excel format.

By 2015, STEVE had been updated several times, and several versions were in use across the country. Several STEVE software components had reached "end-of-life" status, and IT departments were concerned about the security risks that STEVE presented to their networks and were reluctant to continue to host STEVE.¹⁸ In 2015, the NAPHSIS Board decided that an entire STEVE "rebuild" would be necessary, and in September 2015, NAPHSIS held a functional requirements meeting to draft specifications for STEVE 2.0.

NAPHSIS and its members released STEVE 2.0 in late 2016 to provide greater functionality, flexibility, and security than the original STEVE software (STEVE 1.0). Development of STEVE 2.0 cost about \$550,000, and annual support and maintenance costs are expected to cost about \$150,000. Each vital registration jurisdiction contributed \$5,000 to help pay for the development of STEVE 2.0 in 2016; the annual STEVE 2.0 support and maintenance fee for 2017 for each jurisdiction is \$3,500.

STEVE 2.0 performs the same functions as STEVE 1.0 and uses the same IJE layouts. In contrast to STEVE 1.0, STEVE 2.0 uses a "web-based user interface with a cloud-based infrastructure hosted on the Association of Public Health Laboratories' (APHL) APHL Informatics Messaging Services (AIMS) platform to provide states and territories with bi-directional core messaging capabilities." STEVE 2.0 can be operated in manual mode from workstations within the vital records office rather than on a server within the IT department. Jurisdictions can automate sending and receiving data through STEVE 2.0, which must be coordinated with the jurisdiction's IT department to assure the security of the data and the continued operation of the STEVE 2.0 service.

¹⁷ The information from the first section of each layout that is sent to NCHS—as well as to other jurisdictions—is the central component of the Vital Statistics Cooperative Program contract NCHS holds with each jurisdiction.

¹⁸ Many jurisdictions had to apply for exceptions to IT security requirements to keep STEVE in operation. While its software was outmoded, NAPHSIS felt that there was little to no danger of a breach because of the way the software was designed and the components it used. On the other hand, the STEVE software could become unstable if even one jurisdiction had technical trouble, which was usually related to changes made to server environments by state or jurisdiction IT departments and to the frequent need to update security certificates to allow PHINMS to run.

Between late December 2016 and late September 2017, STEVE 2.0 relayed over 66 million messages. Its only significant outages have been the result of reliance on the aging PHINMS system, not due to its own infrastructure. Security certificates are upgraded in one location, thereby relieving jurisdictions of this task, which was one of the limiting factors of STEVE 1.0. In addition, STEVE 2.0 has a modular design, which allows upgrades and enhancements with little or no downtime. Because STEVE 2.0 resides on the AIMS platform, it is able to handle HL7 and other messaging protocols and take advantage of a variety of technical methods for automation.

Table 8. Data items, vital registration jurisdictions, and years of data accessible through six information systems based on vital records that provide data about vital events to various users

	Full DMF	LADMF	EVVE	EVVE FOD	NDI	STEVE
Data item						
Name, First	x	x	x	x	x	NA
Name, Last	x	x	x	x	x	NA
DOB	x	x	x		x	NA
DOD	x	x	x	x	x	NA
Sex			x	x	x	NA
SSN	x	x			x	NA
Birth Certificate #			x			NA
Death Certificate #			x	x	x	NA
Jurisdiction of birth			x			NA
Jurisdiction of death			x	x	x	NA
Cause of death					x	NA
Coverage						
Number of participating U.S. vital registration jurisdictions	57	0	55*	42†	57	57
% U.S. birth records	NA	NA	~87%	NA	NA	NA
% U.S. death records	>98%**	~60%††	~90%	>65%	100%	NA
Years covered	1936– 2017	1936– 2017	1800s– 2017‡	1800s– 2017‡	1979– 2016	NA

NA = Not applicable

* EVVE does not currently include births and deaths from New York State and Texas.

† Number of participating jurisdictions varies by the intended use of the data.

** The author was unable to confirm this estimate of % U.S. death records with SSA. The estimate applies to deaths that occurred after ~2000 and is based on the assumption that virtually all deaths are currently reported to SSA through either jurisdiction-based EDRS, jurisdiction-based data files containing data needed by SSA for the DMF, or copies of death certificates submitted to SSA by jurisdictions unable to send electronic data files.

†† The author was unable to confirm this estimate of % U.S. death records with SSA. The estimate applies to deaths that occurred after ~2011 and is based on the assumption that SSA does not receive data from non-VRO sources for about the 40% of deaths that occur each year in the United States.

‡ The first year for which vital events data are available varies by jurisdiction.

Sources of information for Appendix C:

(1) CDC: NCHS: National Death Index, <https://www.cdc.gov/nchs/ndi/index.htm>; accessed on 9 Oct 2017.

- (2) Department of Commerce, National Technical Information Service, Certification Program for Access to the Death Master File. *Federal Register*, June 1, 2016, Volume 81, Number 105, pages 34882-34895; available at <https://www.gpo.gov/fdsys/pkg/FR-2016-06-01/html/2016-12479.htm>.)
- (3) Department of Commerce, National Technical Information Service. List of persons certified to receive the Limited Access Death Master File as of June 1, 2017, available at <https://classic.ntis.gov/assets/pdf/DMFcertifiedList.docx>; accessed on 4 Oct 2017.
- (4) Email communication with NAPHSIS staff Andrea Price and Bill Bolton on 13 Oct 2017.
- (5) Email communication with NAPHSIS staff Anthony Stout on 16 Oct 2017.
- (6) Email communications with NAPHSIS Executive Director Shawna Webster on 12–13 Dec 2017.
- (7) NAPHSIS: About EVVE FOD, available at <https://www.naphsis.org/evve>; accessed on 4 Oct 2017.
- (8) NAPHSIS: About EVVE FOD: FOD Use Maps. Available at: <https://www.naphsis.org/evve-fod>; accessed on 6 Oct 2017
- (9) NAPHSIS: About EVVE. Available at <https://www.naphsis.org/evve>; accessed on 4 Oct 2017; email communication with NAPHSIS staff Anthony Stout on 16 Oct 2017.
- (10) NAPHSIS: Information systems for vital records stewardship: STEVE: available at <https://www.naphsis.org/systems>, accessed on 6 Oct 2017;
- (11) NAPHSIS: STEVE 2.0: available at <https://www.steve2.org>, accessed on 6 Oct 2017;
- (12) NTIS: Limited Access Death Master File (<https://classic.ntis.gov/products/ssa-dmf/#>; accessed on 4 Oct 2017).
- (13) NTIS: Limited Access Death Master File Available Through Value-Added Online Products, available at <https://classic.ntis.gov/products/ssa-online/>; accessed on 4 Oct 2017.
- (14) Rothwell CJ (CDC/NCHS), SSA Death Master File & NCHS National Death Index: How do they relate? [Slide Presentation], available at <https://www.cdc.gov/nchs/data/bsc/rothwell.pdf>; accessed on 9 Oct 2017.
- (15) Social Security Death Master File, available at www.ssdmf.com/; accessed on 4 Oct 2017.
- (16) Social Security: Compilation of the Social Security Laws: Evidence, Procedure, and Certification for Payment: Sec. 205. [42 U.S.C. 405], available at https://www.ssa.gov/OP_Home/ssact/title02/0205.htm; accessed on 9 Oct 2017.
- (17) Social Security: Requesting The Full Death Master File (DMF), available at https://www.ssa.gov/dataexchange/request_dmf.html; accessed on 17 October 2017.
- (18) U.S. Department of Commerce: National Technical Information Service. Important Notice: Change in Public Death Master File Records. Available at <https://classic.ntis.gov/assets/pdf/import-change-dmf.pdf>; accessed on 4 Oct 2017.
- (19) UberOps, NAPHSIS. STEVE v2.0 High Level Design: available at <https://www.steve2.org/#/resources>, accessed on 6 Oct 2017; and

Appendix D. List of persons certified to receive the Limited Access Death Master File as of June 1, 2017

Source: National Technical Information Service: Limited Access Death Master File: List of persons certified to receive the Limited Access Death Master File, available at <https://classic.ntis.gov/assets/pdf/DMFcertifiedList.docx>; accessed on 9 Oct 2017.

Arizona Department of Transportation-MVD	Apollo Education Group Inc.	District
Abacus Settlements, LLC	Appriss Inc.	Borgess Health
Access2care LLC	AR DFA Individual Income Tax Section	Boston Medical Center
Axiom Corporation	Arizona Health Care Cost Containment System (AHCCCS)	Healthnet Plan
ADP Screening & Selection Svcs	Asset Protection Unit Inc.	Boston Mutual Life Insurance Company
ADP Screening and Selection Svcs	Associated Press	Boston Scientific
AETNA Inc. (Aetna Medicaid Administrators)	Aurora Health Care	Boston University
AIS Recovery Solutions LLC	Automation Research Inc.	Brigham & Women's Hospital
Albany-Schoharie-Schenectady-Saratoga Boces dba Capital Region Boces	Avera Health	Copdgene Study
Alegent Health Creighton Saint Joseph Managed Care Services	Avesis Incorporated & Subsidiaries	Brightpoint
Alliance One International Inc.	BAE Systems, Inc.	Brown & Brown of Pennsylvania, LP
Alphacare of New York	Bank Compensation Consulting	BWW Law Group LLC
Altru Health System	Bank of America Mortgage Data Technology Group	California Insurance Guarantee Association (CAIGA)
American Express	Baylor Heart and Vascular Institute	California State Controller's office Division of Audits
American Family Archives & Chronicles DbA	BCBSM, Inc., D/B/A Blue Cross and Blue Shield of Minnesota	Cambia Health Solutions
Obituarydata.Com	Becklin Research Inc.	Capital Forensics Inc.
American Fidelity Life Ins. Co	Bellmore-Merrick Central High School District	Careington International Corporation
American Health and Life Insurance Company	Benefit Allocation Systems Inc.	Caris MPI, Inc.
American Heritage Federal Credit Union	Benefit Plan Manager	Carpenters Combined Funds Inc.
American Research Bureau	Benevis, LLC	Catalist, LLC
American Underwriters Life Insurance	Beth Israel Deaconess Medical Center	Catholic Life Insurance
Amerigroup	Blue Cross & Blue Shield of Montana	Catholic Relief Services
AMGI Multispecialty Group	Blue Cross & Blue Shield of Texas	CBIZ Benefits & Insurance Services, Inc.
D/B/A Esse Health	Blue Shield of California	CDYNE Corporation
Anchor Computer Inc.	Bluebonnet Life Insurance Co.	Central Research Inc.
Ann Arbor VA Healthcare System (Sub is under Ann Arbor VA Medical Center)	Board of Trustees of Leland Stanford Junior University, On Behalf of Its School of Medicine	Central States Health and Life Co. of Omaha
Anselmo Lindberg Oliver LLC	Boetger & Associates Inc. & Northwest Savings Bank	Central States Indemnity Co of Omaha and CSI Life
Anthem, Inc.	Boone County Fire Protection	Central Texas Community Health Centers DBA Communitycare
Aperture Credentialing LLC		Chan Healthcare, A Subsidiary of Crowe Horwath LLP

Charlotte-Mecklenburg Hospital Authority AKA Carolinas Healthcare System	Corelogic Solutions, LLC	Examone
Chatham County Government Children's Hospital of Philadelphia	Cornerstone Education Loan Services	Experian Health Inc.
Chronic Disease Research Group (CDRG)	CreDENTIALS Services Inc. dba Verifpoint/Credentialing Solutions	Experian Marketing Services
CIGNA (LINA) (PA)	Credit Shop Inc.	Express Scripts Holding Company
CIGNA Health & Life Insurance Co. On Behalf of Self and Subsidiaries	Cross Country Computer Corp	Extend Health Inc., Willis Towers Watson
City of Bristol	Crump Life Insurance Services	Fair Isaac Corporation
City of Buffalo	D Greene & Company	Fannie Mae
City of Hoboken	Data Center Inc. (DCI)	Farm Bureau Life Insurance Company of Missouri
City of Niagara Falls, NY	Datalab USA	Fast Data Services
City of Rochester	Dataline Inc.	Federation of State Medical Boards
City of Sanibel	DCM Services LLC	FHC of Puerto Rico Inc.
Clarity Services Inc.	DCS Information Systems	Fiduciary Trust Company International
Clinical Reference Laboratory, Inc.	Dearborn National Life Insurance Company of New York	Financial Industry Regulatory Authority Inc. (FINRA)
Colorado Department of Revenue	Delta Dental of Puerto Rico, Inc.	First American Financial Corporation
Columbia University	Department of Community Services & Development (HEAP)	First County Bank
Columbia Valley Community Health	Dillard's Inc.	Flagstar Bank
Columbian Mutual Life Insurance Company	Discover Bank	Flatiron Health Inc.
Comerica Bank	Discover Products, Inc.	Florida Department of Corrections
Commonwealth of Massachusetts, Office of State Auditor	DISH Network, LLC	Florida Department of Highway Safety and Motor Vehicles
Commonwealth of Pa, Insurance Dept., Bureau of Special Funds, Workers' Comp Security Fund, Auto Cat Fund	Doyle & Foutty, PC	Florida Department of State
Community Care Network of VA Comprehensive Health Management - WellCare	DST Systems, Inc.	Florida Hospital Orlando
Computer Services, Inc.	Duke Cardiovascular Magnetic Resonance Center	Foresters Life Insurance and Annuity Company
Concordia Plan Services	Duke Patient Revenue Management Organization	Franchise Tax Board
Confluence Health	Early Warning Services LLC	Franciscan Health System
Consol Energy Inc. - CNX Land LLC	East Syracuse Minoa Central School District	Freeman Physician Hospital Organization - FPHO - Freeman Health System
Cookeville Regional Medical Center	Echo, Inc.	Fresenius Medical Care, North America
Coordinated Care Corporation dba Managed Health Services	EECS LLC	Fuzion Analytics Inc.
	Electronic Medical Office Logistics (EMOL)	Gabriel Roeder Smith & Company
	Emblem Health	Garretson Resolution Group
	Employment Screening Services Inc.	Geisel School of Medicine at Dartmouth College, Dartmouth-Hitchcock Medical Center
	Equitable Life & Casualty Insurance Company	Geisinger Clinic
	Essen Medical Associates PC	General Board of Pensions
	Essentia Health	
	Estate Information Services, LLC	
	Everyone Counts Inc.	
	Evolution Consulting LLC	

General Information Services Inc.	HMS Inc. - Health Management Systems Inc.	Kansas Department of Revenue
Genworth Financial	HS1 Medical Management Inc.	Kansas Department of Revenue - State of Kansas
Georgia Department of Revenue	HubHealth of South Mississippi, Inc.	Karmadata, Inc.
GMG Health Systems Associates, Pa dba, Gonzaba Medical Group	Humana Inc.	Keck Medicine of USC
Golden Gate Bridge, Highway & Transportation District	Huntington Hospital	Kelmar Associates
Gorra Childrens Trust	HygenicsData LLC	Keyport Investors, LLC
Great Lakes Educational Loan Services	Icahn School of Medicine at Mount Sinai	Kinnect Services
Guardian Research Network Inc.	ID Analytics, LLC	L2 Inc.
Guidestone Financial Resources	Idaho Transportation Department, Division of Motor Vehicles	Labor-Management Healthcare Fund
Gundersen Lutheran Admin Svcs. Inc.	Idology Inc.	Lake Huron PHO
Hannis T. Bourgeois, LLP	Illinois Dept of Healthcare & Family Services Office of Inspector General	Layline Energy, LLC Formerly Known as Layline Petroleum, LLC
Hart Software, Inc.	Illinois Office of The Auditor General	Leadership Institute
Harvard University Institute for Quantitative Social Science	Imc Salud - First Medical Health Plan Inc.	LeadsOnline LLC
Hattiesburg Clinic, PA	Impac Medical Systems, an Elekta Company	Legacy Health
Hawaii Medical Service Administration (Hmsa)	Impact Makers Inc.	Lexis Nexis Risk Data Retrieval Services LLC
Headco Industries, Inc.	Indecs Corporation	Life Assurance Fund Operating Company
Health Network Solutions Inc.	Infutor Data Solutions LLC	Life Status 360, LLC
Health One Alliance	Inogen, Inc.	Lifepans, Inc.
Health Plan of The Upper Ohio Valley Inc.	Inova Health System	Loews Corporation
Healthcore Inc.	Inspira	Louisiana Department of Revenue
Healthfirst Management Services, LLC	Institute for Fair Elections	LTC Partners
Healthmarkets Inc.	Integrated Solutions Health Network	Lutheran Church-Missouri Synod Foundation (LCMS Foundation)
Healthpartners Inc.	Intellicorp Records Inc.	M&A Profilers, Inc. dba Colley Investigations
Healthplex	Interactive Data LLC	M2s, Inc.
Heart Imaging Technologies LLC, A Delaware Corporation	Interlife America Inc.	Maine Revenue Services
Hennepin County Medical Center	International Union, UAW	Mary's Center
Heritage Pension Advisors LLC	Invenio, LLC	Massachusetts Department of Revenue
Heritage Provider Network	Ivision International of Puerto Rico Inc.	Massachusetts Department of Unemployment Assistance
Hewlett Packard Enterprise (OH)	Jackson Clinic, PA	Massachusetts Mutual Life Insurance Company, C.M. Life Insurance Company, MML Bay State Life Insurance Company, And MML Investors Services LLC
Hewlett Packard Enterprise (WI)	John Hancock Life Insurance Company USA (John Hancock Mutual Life Ins. Co)	MassDOT Registry of Motor Vehicles
Highlands Wellmont Health Network	John Hopkins Hospital	Maximus Health Services, Inc.
HME Recovery, LLC	Johns Hopkins Bloomberg School of Public Health	
	JP Morgan Chase	

McDermott Will & Emery LLP	Multiplan Inc.	Northshore University
McKesson Specialty Health	Multiple Injury Trust Fund	Healthsystem
MDeverywhere Inc.	Muscogee (Creek) National	Nova Management Services Inc.
Medica Health Plans	Citizenship office	NY State Dept. of Tax & Finance
Medical Provider Resources	National Benefit Life	(New York State Department of
Medimore, Inc.	National Council of The State	Taxation and Finance)
Medtronic PLC	Boards of Nursing (NCSBN)	NYS Gaming Commission
Medversant Technologies LLC	National Jewish Health	Office of The Chief Financial
Melissa Data Corporation	National Medtrans Network	officer, Office of Tax & Revenue
Memorial Health	Naveos LLC	(Washington, DC)
Services/Memorial Health Care	Navigant Consulting Inc.	Office of The Texas Secretary of
System	Navion Healthcare Solutions,	State
Memorialcare Health System	LLC	Ohio Department of Taxation
Menno Haven Inc.	NCAN Services, Inc.	Ohio Public Employees
Mercy	Nebraska Health Partners	Retirement System
Mercy Medical Center - Des	Neighborhood Health Plan of	Ohio State University Wexner
Moines	Rhode Island	Medical Center
Mercy of Iowa City Regional	Nelnet Service, LLC	Oklahoma Tax Commission
Physician Hospital Organization	Nelnet Servicing, LLC	Olmsted Medical Center
(PHO)	Network Solutions IPA	Onco Inc.
Methodist Health Partners	New Jersey Division of Taxation	One Source Technology dba
Metlife	New Mexico Taxation And	Asurint
Metro Plus Health Plan	Revenue Department	OPENonline, LLC
Metrocare Physicians	New York City Department of	Optum
Metropolitan Health Plan	Housing Preservation And	Optuminsight, Inc. (KS)
Mhmr of Tarrant County	Development (HPD NYC)	Optumrx
Mib Solutions Inc.	New York Methodist Hospital	OSI Vision, LLC
Michael Moecker & Associates	New York State Catholic Health	Ozark National Life Insurance
Michigan (Mi) Department of	Plan Inc. (D/B/A Fidelis Care)	Company
HLth & Human Services	New York State Office of The	Painters Union Pension Fund
Michigan Department of State	State Comptroller - Office of	Palladian Health LLC
Microbilt Corporation	Unclaimed Funds	Partners Healthcare System Inc.
Midwest Employers Casualty	New York State Office of The	Peacehealth CVO
Company	State Comptroller, New York	Pediatric Associates Saltzman,
Milliman Inc.	State and Local Retirement	Tanis, Pittel, Levin
Minnesota Department of	System	Penn Medicine
Human Services	New York State Teachers'	Pennsylvania Higher Education
Mission Hospital, Inc.	Retirement System	Assistance Agency (PHEAA)
Mississippi Department of	New York-Presbyterian Hospital	Pension Benefit Information
Revenue (Ms Dept of Revenue)	NJ Division of Family	Inc.
Missouri Secretary of State	Development, Dept. of Human	Phin Solutions Inc.
Modesto Irrigation District	Services	Pinnacle Care Inc.
Mohela	NJ Motor Vehicle Commission	Plainedge UFSD
Molina Medicaid Solutions	North Carolina Department of	PNT Data Corp
Monroe 2-Orleans BOCES	Information Technology	Pointright Inc.
Montefiore Information	North Mississippi Health Link,	Pointserv Technologies, LLC
Technology	Inc.	Polish National Alliance of The
Multicare Health System	Northeast Escrow Services LLC	U.S. of North America

Political Data Inc.	Group	State of Delaware, Division of Revenue
Pondera Solutions, Inc.	Savannah College of Art and Design	State of Illinois Department of Revenue
Port Authority of New York (NY) & New Jersey (NJ)	Schoedinger Funeral Service	State of Michigan, Department of Treasury, Discovery & Tax Enforcement
PRA Receivables Management LLC	Scion Dental	State of Rhode Island Department of Human Services
Preston Ventures LLC	Screen Actors Guild Producers Pension Plan - SAG Pension Health Plan	State of Tennessee HCFA Bureau of TENNCARE
Pricewaterhousecoopers	Scripps Health	Stern & Eisenberg PC
Primerica Life Insurance Company	SE2, LLC	Stony Brook University Hospital
Primewest Health	Seattle Children's Hospital, Medical Staff Services	Stony Brook University of School Medicine
Principial Financial Group	Securecare Inc.	Streamline Verify
Probate Recovery Systems LLC	Securian Financial Group Inc.	Subsequent Injury Fund
Professional Credential Verification Service Inc. (PCVS)	Security Mutual Life Insurance Company of New York	Sumo Group Inc.
Progressive Data Services	Security Software Solutions	SUNY College of Optometry/University Eye Center
Providence Health and Services	Select Medical	SUNY Upstate University Hospital
Prudential	Sensient Technologies Corporation	Surety Life & Casualty Insurance Company
PSC-CUNY Welfare Fund	Sentara Healthcare	Sutherland Healthcare Solutions, Inc.
P-Solve, LLC	Sentinel Security Life Insurance Service Corporation	Sutter Health
Public Health - Seattle & King County	International (SCI)	Sutter Shared Services
Q Capital Strategies LLC	Seton Insurance Services	Swedish Health Services
Quantum3 Group LLC	Sharp Healthcare	Technosoft Corporation
R L Polk & Co	Signature Bank	Tenet Healthcare Corporation
Rapidcourt LLC	Simpson Thacher & Bartlett, LLP	Tennessee Department of Correction
Regenstrief Institute, Inc.	Skagit County Public Hospital, District 1	Texas Vascular Associates
Regional Justice Information Service	Slome Capital Corp	The Advisory Board Company
Reliance Standard Life Insurance Company	South Carolina PEBA	The Berwyn Group Inc.
Ri Executive office of Health & Human Services	South Carolina State Election Commission	The Bessemer Group, Inc.
Riversource Life Insurance Company and Riversource Life Insurance Co. of New York	Southeastrans	The Brooklyn Hospital Center
Rush University Medical Center	Southern Financial Life Insurance Co	The Everett Clinic
Rutgers, The State University of New Jersey, office of Enterprise Risk Management	Southwest Medical IPA Alliance	The Guthrie Clinic
Saint Francis Health System	SSDC Services	The Ministers and Missionaries Benefit Board (MMBB)
Saint Joseph Mercy Health System (sub under St Joseph Mercy Hospital)	St Barnabas Hospital	The Polyclinic
Sallie Mae Bank	St Jude Medical, Inc.	The Verification Group
Sandhills Center for MH/DD/SAS	Stanislaus County District Attorney's office	Thomson Reuters
Santa Barbara Tax Products	State Bar of Texas	TINCheck, LLC
	State Compensation Insurance Fund	Toolcase LLC
	State Farm Life Insurance Companies	

Tougaloo College	University of Massachusetts	Owned Subsidiary Advanced
Transamerica Life Insurance	Medical School	Resolution Services, Inc. (Vasa
Company and Its Affiliates	University of Michigan Health	Inc.)
Transunion LLC	System	Visions Federal Credit Union
Transunion Risk and Alternative	University of Mississippi	Vital Decisions, LLC
Data Solutions, Inc.	Medical Center	Wake Forest Baptist Medical
Trico Products Corporation	University of North Carolina At	Center
Triple-S Salud, Inc.	Chapel Hill, Collaborative	Walla Walla Clinic
Triplus Services Inc.	Studies Coordinating Center	Washington (WA) State Social
Truven Health Analytics, An IBM	University of Pittsburgh	and Health Services
Company	University of Texas	Washington State Board for
Typhoon Data, LC	Southwestern Medical Center	Volunteer Firefighters And
UAB Transplant Informatics	University of Vermont Health	Reserve officers
UAMS - MIRT (Board of	Network (Credentialing &	Washington State Department
Trustees of The University of	Enrollment Department)	of Labor and Industries
Arkansas Acting for and on	University of Vermont Medical	Wayne State University
Behalf of The University of	Center	Physician Group
Arkansas For Medical Sciences)	University of Virginia	WEA Insurance Corporation
UC San Diego Health	University of Washington	Welland Company
Udren Law offices, Pc	School of Medicine (UW SOM)	Wellplace, MI
UMC Physicians	office of Medical Staff	West Virginia Clinical and
UMWA Health & Retirement	Appointments	Translational Science Institute
Funds	University Physicians'	(WVCTSI)
UNCAS & Company	Association Inc.	West Virginia United Insurance
Universal Life Insurance	University Physicians Network	Services, Inc./Health Partners
Company	LLC	Network, Inc.
Universal Patient Key Inc.	Unum Group	Westchester Management
University (Univ) of Wisconsin	UPCR LLC	Services Organization, LLC
(WI) School of Medicine	UPMC	Western Psychological and
University Clinical Health	Usable Life Insurance Company	Counseling Services Pc
University Health Shreveport	UTG Inc.	Western Union
University Health System	UW Medicine Valley Medical	Westmed Practice Partners
University of California San	Center	Westside Family Healthcare Inc.
Francisco (UCSF) Interstitial	Valence Health	Wheaton Franciscan
Lung Disease Program	VCU Health System	Healthcare-Iowa, Inc. D/B/A
University of Chicago	Venio LLC DbA Keane	Covenant Centralized
University of Iowa Clinical Staff	Verafin Inc.	Credentials Verification
office	Verato, Inc.	Organization
University of Kansas Hospital	Verisk Analytics, Inc.	WI Insurance Security Fund
University of Kansas Medical	Verisys Corporation	Will Notification LLC
Center - Greater Plains	Verizon Communications	Willis-Knighton Health System
Collaborative Clinical Data	Verus Financial LLC	Wisconsin Department of
Research Network	Virginia Department of	Revenue
University of Maryland	Elections	Xerox State & Local Solutions
Baltimore, Department of	Virtua	Inc.
Anesthesiology	Visa Inc., including Its Wholly	Zebu Compliance Solutions

Appendix E. Cost estimate for Vital Records Systems in the United States

(1) Estimated population of 10 states in 2013: 65,143,263 (10 states are AL, FL, IL, KS, OK, MD, MI, ND, OR, and VT)

(Source: <https://www.census.gov/data/tables/2016/demo/popest/state-total.html>)

(2) Estimated population of New York City in 2013: 8,405,837

(Source:

https://factfinder.census.gov/faces/tables/services/jsf/pages/productview.xhtml?pid=PEP_2014_PEPANNRES&prodType=table)

(3) Total estimated population of for 10 states and NYC in 2013: $65,143,263 + 8,405,837 = 73,549,100$

(4) Estimated cost of Vital Records System for 10 states and NYC in 2013: $2 \times \$16,624,274 = \$33,248,548$

(Estimate is based on assumptions that these 10 states and NYC are representative of the United States as a whole, and that SSA payments for Enumeration at birth (EAB) to these 11 jurisdictions, i.e., \$3,324,855, is 20% of the cost for their birth systems and that the cost of their death systems is the same as the cost for birth systems.)*

(5) Estimated population of United States in 2013: 316,204,908

(Source: <https://www.census.gov/data/tables/2016/demo/popest/state-total.html>)

(6) Proportion of U.S. population in 10 states and NYC in 2013: $73,549,100/316,204,908 = 0.232599489$

(7) Estimated cost of Vital Records Systems for United States in 2013: $\$33,248,548/0.232599489 = \$142,943,341$.

(8) Estimated cost per capita for Vital Records Systems for United States in 2013:

$\$33,248,548/73,549,100 = \0.452059209 .

**Estimate based on information gathered by Jennifer Woodward, Manager of the Oregon Center for Health Statistics, and NAPHSIS in 2013 as background for cost estimates for vital records.*

Appendix F. Availability and cost of birth, death, and fetal death certificates and number of births, deaths, and fetal deaths registered, by vital registration jurisdiction, United States, 2016.

Jurisdiction	Availability of certificates		Cost of certificates				Number of events, 2016		
			Birth		Death				
	Birth	Death	First	Additional	First	Additional	Births	Deaths	Fetal Deaths
Alabama	Closed	Closed	\$15.00	\$6.00	\$15.00	\$6.00	57,647	51,431	500
Alaska	Closed	Closed	\$30.00	\$25.00	\$30.00	\$25.00	11,095	4,355	74
American Samoa	Closed	Closed	\$5.00	\$5.00	\$5.00	\$5.00	1,077	281	12
Arizona	Closed	Closed	\$20.00	\$20.00	\$20.00	\$20.00	85,573	57,215	527
Arkansas	Closed	Closed	\$12.00	\$12.00	\$10.00	\$8.00	36,913	31,028	207
California	Open*	Open	\$25.00	\$25.00	\$21.00	\$21.00	489,976	263,201	2,400
Colorado	Closed	Closed	\$17.75	\$10.00	\$20.00	\$13.00	67,183	38,151	396
Connecticut	Closed	Open	\$30.00	\$15.00	\$20.00	\$20.00	37,467	30,641	172
Delaware	Closed	Closed	\$25.00	\$25.00	\$25.00	\$25.00	11,415	8,980	72
District of Columbia	Closed	Closed	\$23.00	\$23.00	\$18.00	\$18.00	14,847	6,055	157
Florida	Closed	Closed*	\$14.00	\$14.00	\$5.00	\$4.00	225,260	199,964	1,627
Georgia	Closed	Open	\$25.00	\$5.00	\$25.00	\$5.00	130,961	81,783	8,304
Guam	Closed	Closed	\$5.00	\$5.00	\$5.00	\$5.00	3,433	1,021	35
Hawaii	Closed	Closed	\$10.00	\$4.00	\$10.00	\$4.00	18,057	11,041	814
Idaho	Closed	Closed	\$16.00	\$16.00	\$16.00	\$16.00	22,179	13,297	128
Illinois	Closed	Closed	\$10.00	\$2.00	\$19.00	\$4.00	150,789	104,092	944
Indiana	Closed	Open*	\$10.00	\$4.00	\$8.00	\$4.00	83,983	64,051	569
Iowa	Open	Open	\$20.00	\$20.00	\$20.00	\$20.00	39,094	29,318	174
Kansas	Closed	Closed	\$15.00	\$15.00	\$15.00	\$15.00	39,317	25,489	195
Kentucky	Open	Open	\$10.00	\$10.00	\$6.00	\$6.00	53,062	47,606	345
Louisiana	Closed	Closed	\$15.00	\$15.00	\$7.00	\$7.00	63,277	44,411	368
Maine	Closed	Closed*	\$15.00	\$6.00	\$15.00	\$6.00	12,477	14,069	55
Maryland	Closed	Closed	\$10.00	\$10.00	\$10.00	\$12.00	73,544	47,235	526
Massachusetts	Open*	Open	\$20.00	\$20.00	\$20.00	\$20.00	71,935	57,861	372
Michigan	Closed	Open	\$34.00	\$16.00	\$34.00	\$16.00	112,340	98,840	623
Minnesota	Closed	Open	\$26.00	\$19.00	\$13.00	\$6.00	69,831	43,134	460
Mississippi	Closed	Closed	\$15.00	\$5.00	\$15.00	\$5.00	57,139	31,016	410
Missouri	Closed	Closed*	\$15.00	\$15.00	\$13.00	\$10.00	75,864	61,888	456
Montana	Open*	Open	\$12.00	\$5.00	\$15.00	\$8.00	12,280	9,941	55
Nebraska	Closed		\$17.00	\$17.00	\$16.00	\$16.00	27,101	16,400	138
Nevada	Open*	Closed	\$20.00	\$20.00	\$20.00	\$20.00	35,915	24,692	230
New Hampshire	Closed	Closed	\$15.00	\$10.00	\$15.00	\$10.00	12,350	11,909	56
New Jersey	Closed	Closed	\$25.00	\$2.00	\$25.00	\$2.00	100,379	72,082	667
New Mexico	Closed	Closed	\$10.00	\$10.00	\$5.00	\$5.00	23,322	177,632	98
New York City	Closed	Closed	\$15.00	\$15.00	\$15.00	\$15.00	120,367	54,280	12,068
New York State	Closed	Closed	\$30.00	\$15.00	\$30.00	\$15.00	114,494	98,766	5,172
North Carolina	Open	Open	\$24.00	\$15.00	\$24.00	\$15.00	122,780	91,349	933
North Dakota	Closed		\$7.00	\$4.00	\$5.00	\$2.00	13,035	6,806	76
Northern Marianas	Closed	Closed	\$20.00	\$20.00	\$15.00	\$15.00	1,217	222	11
Ohio	Open	Open	\$21.50	\$21.50	\$21.50	\$21.50	138,570	119,618	893
Oklahoma	Closed*	Closed*	\$15.00	\$15.00	\$15.00	\$15.00	51,319	38,170	308
Oregon	Closed	Closed	\$25.00	\$20.00	\$25.00	\$20.00	45,533	35,799	186
Pennsylvania	Closed	Closed	\$20.00	\$20.00	\$9.00	\$9.00	138,637	134,523	1,257

Jurisdiction	Availability of certificates		Cost of certificates				Number of events, 2016		
			Birth		Death				
	Birth	Death	First	Additional	First	Additional	Births	Deaths	Fetal Deaths
Puerto Rico	Closed	Closed	\$5.00	\$10.00	\$10.00	\$10.00	28,326	29,626	419
Rhode Island	Closed	Closed	\$20.00	\$15.00	\$20.00	\$15.00	11,430	10,023	697
South Carolina	Closed	Closed	\$12.00	\$3.00	\$12.00	\$3.00	53,810	47,123	466
South Dakota	Closed	Closed*	\$15.00	\$15.00	\$15.00	\$15.00	12,910	7,945	88
Tennessee	Open*		\$15.00	\$15.00	\$15.00	\$15.00	86,540	71,507	631
Texas	Open	Closed*	\$22.00	\$22.00	\$20.00	\$3.00	406,945	194,845	2,263
Utah	Closed	Closed*	\$20.00	\$8.00	\$18.00	\$8.00	52,121	18,499	472
Vermont	Open	Open	\$10.00	\$10.00	\$10.00	\$10.00	5,567	5,787	24
Virgin Islands	Closed	Closed	\$12.00	\$12.00	\$12.00	\$12.00	1,197	651	32
Virginia	Closed	Closed	\$12.00	\$12.00	\$12.00	\$12.00	101,216	65,580	4,654
Washington	Open*	Open	\$20.00	\$20.00	\$20.00	\$20.00	90,301	54,767	562
West Virginia	Closed	Closed	\$12.00	\$12.00	\$12.00	\$12.00	19,887	21,990	98
Wisconsin	Closed	Open	\$20.00	\$3.00	\$20.00	\$3.00	66,239	51,650	324
Wyoming	Closed	Closed	\$13.00	\$13.00	\$10.00	\$10.00	6,710	4,347	32
Total							4,016,233	2,943,983	53,832
Minimum			\$5.00	\$2.00	\$5.00	\$2.00	1,077	222	11
Median			\$15.00	\$15.00	\$15.00	\$11.00	51,720	36,975	370
Maximum			\$34.00	\$25.00	\$34.00	\$25.00	489,976	263,201	12,068

* The availability of the birth or death certificate is restricted to individuals with a direct relationship to the individual named on the certificate, or there are other requirements that must be met to obtain a copy of a certificate. These restrictions and requirements may vary by jurisdiction.

Sources of information in Appendix F:

- (1) *Emails from Andrea Price, NAPHSIS, on 17 and 18 October 2017.*
- (2) *Extract of NAPHSIS database on jurisdictions, provided by Andrea Price, NAPHSIS on 17 October 2017.*
- (3) *Numbers of events for 2016 are from the NAPHSIS database on jurisdictions; these numbers are derived from NCHS/NVSS reports for natality and mortality events received through the Vital Statistics Cooperative Program.*

Appendix G. NCHS Vital Statistics Cooperative Program funding awarded for special projects, by vital registration jurisdiction, United States, 2013–2017.

Jurisdiction	Total Special Project Funding Awarded in 2013-2017
Alabama	\$8,500.00
Alaska	\$6,000.00
American Samoa	\$6,000.00
Arizona	\$8,500.00
Arkansas	\$426,350.00
California	\$644,810.57
Colorado	\$8,000.00
Connecticut	\$8,500.00
Delaware	\$6,000.00
District of Columbia	\$416,915.20
Florida	\$484,696.90
Georgia	\$218,500.00
Guam	\$6,000.00
Hawaii	\$473,700.00
Idaho	\$255,818.00
Illinois	\$9,000.00
Indiana	\$438,500.00
Iowa	\$689,037.00
Kansas	\$152,563.00
Kentucky	\$8,000.00
Louisiana	\$214,713.00
Maine	\$155,999.84
Maryland	\$178,095.00
Massachusetts	\$369,614.00
Michigan	\$214,719.00
Minnesota	\$218,000.00
Mississippi	\$8,500.00
Missouri	\$218,500.00
Montana	\$6,000.00
Nebraska	\$217,000.00
Nevada	\$342,828.44
New Hampshire	\$436,000.00
New Jersey	\$258,500.00
New Mexico	\$562,500.00
New York City	\$588,879.00
New York State	\$218,099.00
North Carolina	\$228,500.00
North Dakota	\$6,000.00
Northern Marianas	\$6,000.00
Ohio	\$8,500.00
Oklahoma	\$187,649.00
Oregon	\$212,851.00
Pennsylvania	\$219,000.00
Puerto Rico	\$6,000.00
Rhode Island	\$6,000.00

Jurisdiction	Total Special Project Funding Awarded in 2013-2017
South Carolina	\$8,500.00
South Dakota	\$8,000.00
Tennessee	\$8,000.00
Texas	\$8,500.00
Utah	\$326,417.71
Vermont	\$135,920.00
Virgin Islands	\$6,000.00
Virginia	\$218,500.00
Washington	\$614,925.00
West Virginia	\$100,806.00
Wisconsin	\$8,000.00
Wyoming	\$155,562.34
Total	\$10,962,969.00
Minimum	\$6,000.00
Median	\$178,095.00
Maximum	\$689,037.00

Source of information in Appendix G: VSCP Summary of 2013-2017 VSCP Special Project Funding, provided by Delton Atkinson, NCHS, on 27 Oct 2017.